

1 The ABC of Binding Theory

1.1 Preliminaries

1.1.1 Reference, coreference, and indexing

What is Binding Theory (BT) about? To a first approximation, BT restricts the distribution of NPs (or DPs, if you prefer) that have the same *referent* (starting with chapter 4, we will add non-referential NPs to the picture, which will be ignored until then). We will indicate sameness of reference, *coreference* for short, by *coindexing*; that is, coreferent NPs carry the same *index*, for which we use integers throughout. Thus in (1.1), the NP *the baroness* and the NP *she* are coindexed, which signals that they are coreferent, which in turn means that they have the same referent – they refer to the same person or thing – namely the actual baroness in flesh and blood:

(1.1) After [_{NP} the baroness]₁ had visited the lord, [_{NP} she]₁ left the house.

Note that on this understanding, BT is relevant for nominal categories only, and only for the maximal projections, i.e. NPs.¹ As a convention we assume that two NPs corefer if *and only if* (iff) they are coindexed. Contra-indexing (or lack of an index on either NP) indicates non-coreference. This is illustrated in (1.2):

- (1.2) (a) After [_{NP} the baroness]₂ had visited the lord, she₂ left the house.
(she=the baroness)
(b) After [_{NP} the baroness]₁ had visited the lord, [_{NP} she]₂ left the house.
(she≠the baroness)

It should be noted that the actual choice of integer is irrelevant; (1.1) expresses the same *coreference pattern* as (1.2a) (as would any sentence in which both occurrences of the index are replaced by the *same* integer). An NP marked 1 is in no sense prior, higher, or superior to one marked 2. All that matters is which NPs have the same index, and which do not.

¹ The latter aspect I consider a genuine fact about Binding Theory. On the view pursued here, indexing on non-maximal projections (e.g. signalling specifier-head agreement or head-movement dependencies) simply is not subject to Binding Theory and should be kept separate from it. As for the former aspects, though there are sentential and adverbial (i.e. PP-) anaphors, little work on their distribution has been done, and we will ignore them here (see e.g. Hegarty *et al.* [2001] and the references therein).

In traditional grammars, the NP *the baroness* in (1.1) is referred to as the *antecedent* of the pronoun *she*. We adopt the following:

- (1.3) Definition: Antecedent
 A is the *antecedent* of B iff (if and only if) (i) A precedes B, and (ii) A and B corefer.

By our convention, an NP will be coindexed with its antecedent (if it has one). This holds for coreferring NPs within a single sentence, and across sentences. The latter, however, are usually not subject to Binding Conditions of the sort discussed here.²

1.1.2 The basic data

Restricting our attention to singular NPs for the time being, two NPs in a given sentence will show one of three logically possible coreference relations (Reinhart, 1983a: 29):

- (1.4) (a) *obligatory coreference*: Zelda bores herself.
 (b) *obligatory non-coreference*: She adores Zelda's teachers.
 (c) *optional coreference*: Zelda adores her teachers.

Given what was said before, grammatical representations for these will look like in (1.5):

- (1.5) (a) $Zelda_1$ bores herself₁.
 (b) She_8 adores $Zelda_{15}$'s teachers.
 (c) $Zelda_4$ adores her₄ teachers. *or*
 $Zelda_4$ adores her₇ teachers.

Ungrammatical representations for (1.4a) and (1.4b) are given in (1.6):

- (1.6) (a) $*Zelda_1$ bores herself₂.
 (b) $*She_8$ adores $Zelda_8$'s teachers.

It will be convenient to summarize patterns as in (1.5) and (1.6) as shown in (1.7), whose logic should be transparent:

- (1.7) (a) $Zelda_1$ bores herself_{1/*2}.
 (b) She_8 adores $Zelda_{15/*8}$'s teachers.
 (c) $Zelda_4$ adores her_{4/7} teachers.

The key insight captured in BT is that the (un)availability of coreference between two NPs crucially depends on two factors:

² See e.g. Grosz *et al.* (1995); Gundel *et al.* (1993); Walker *et al.* (1998) and the references therein for some discussion of trans-sentential anaphora.

- the morphological shape of the NPs
- the structural relation between the NPs

This is not meant to exclude the possibility of additional factors that influence coreference options (which will be discussed especially in chapters 3 and 11). First, however, we will introduce the relevant NP-types of English and then, in turn, explore and characterize the syntactic configurations in which they require, allow, or disallow coreference.

1.1.3 Three types of NPs

Virtually all approaches to BT in English distinguish three types of NPs by (mostly) morphosyntactic criteria. These are illustrated in (1.8a–1.8c):

- (1.8)
- (a) reflexives and reciprocals ('anaphors'):
 - himself, herself, itself, themselves, myself, yourself, ourselves, yourselves
 - each other, one another
 - (b) non-reflexive pronouns ('pronominals'):
 - he, she, it, him, her, I, us, you, me, his, your, my, our
 - (c) full NPs including names ('r-expressions'):
 - the baroness, Peter, this, a disinherited Russian countess . . .

In parentheses I have given the terms for these categories as used in the influential work of Chomsky (e.g. 1981) and his school: anaphor, pronominal, and r-expression (with *r* reminiscent of 'referential'). For the first two, a cautionary remark is in order, because they unfortunately provide potential for confusion: traditionally the term *anaphor* (often with the plural *anaphors* rather than *anaphora*) is used for any NP, reflexive or not, that has an antecedent. Likewise, the term *pronominal* invites confusion with the traditional notion of *pronoun*, which applies to reflexive and non-reflexive pronouns alike. We will thus stick to the terms 'reflexive/reciprocal', 'non-reflexive pronoun', and 'full NP' in the remainder of this book.

We will now motivate this tripartition, starting with reflexives versus the rest (reciprocals, being necessarily plural, will not be discussed until chapter 10). Consider the sentences in (1.9):

- (1.9)
- (a) That it rains bothers Peter.
 - (b) That it rains bothers her/him.
 - (c) *That it rains bothers himself/herself.

All these sentences contain but one referential NP (the expletive *it* is of no interest to BT, since it lacks a referent – and perhaps semantic content in general). We can thus omit the indexing for expository convenience, given that no coreference is involved. We simply observe that reflexives cannot occur in this configuration, while both non-reflexive pronouns and full NPs can.

Table 1.1 *Distribution of the three NP-types*

configuration	ex.	reflexive	non-reflexive	full NP
no antecedent	(1.9)	*	ok	ok
non-local antecedent	(1.11)	*	ok	*
local antecedent	(1.10)	ok	*	*

Inversely, only reflexives, but neither non-reflexives nor full NPs, are permitted in (1.10):

- (1.10) (a) *Peter₃ watches Peter₃ in the mirror.
 (b) *Peter₃ watches him₃ in the mirror.
 (c) Peter₃ watches himself₃ in the mirror.

(Note that the two occurrences of *Peter* in [1.10a] are coindexed, indicating that we speak about the same Peter. The sentence is presumably acceptable if I point at a different Peter upon using the names, just as [1.10b] is of course grammatical if the pronoun is not coindexed with the name.)

Let us finally turn to the difference between non-reflexive pronouns and the rest, illustrated by way of the sentences in (1.11):

- (1.11) (a) *Carla₄ thinks that I hate Carla₄.
 (b) Carla₄ thinks that I hate her₄.
 (c) *Carla₄ thinks that I hate herself₄.

Here, reflexives pattern with full NPs, and in contradistinction to non-reflexive pronouns. Note that the difference between (1.10) and (1.11) is not the absence versus presence of an antecedent (there is one in each), but seems to be one of syntactic *locality*: the antecedent NP is within the same clause as the anaphor in (1.10), but in a higher clause in (1.11). We summarize these (preliminary) results in table 1.1. What is clear from this table is that at least this three-way distinction needs to be recognized to distinguish correctly the coreference options of NPs in English. Notice also that reflexive and non-reflexive pronouns seem to be in complementary distribution. We will now characterize the conditions for coreference for the three types of NPs in turn.

1.2 Binding

1.2.1 Reflexive and non-reflexive pronouns

We observed above that reflexive pronouns require an antecedent, and an antecedent within their local clause at that. This is illustrated in more detail in (1.12):

- (1.12) (a) *That it rains bothers himself/herself. (no antecedent)
 (b) *Carla₄ thinks that I hate herself₄. (non-local antecedent)
 (c) Peter₂ watches himself₂ in the mirror. (local antecedent)

Turning now to non-reflexive pronouns, recall that they can occur with or without a sentence-internal antecedent, cf. (1.13), as long as the antecedent is not in the same local clause, cf. (1.13c):

- (1.13) (a) That it rains bothers him/her. (no antecedent)
 (b) Carla₄ thinks that I hate her₄. (non-local antecedent)
 (c) *Peter₃ watches him₃ in the mirror. (local antecedent)

Based on these data we formulate our first version of the *Binding Conditions*:

- (1.14) Binding Conditions (preliminary)
 (A) A reflexive pronoun must have an antecedent within its local clause.
 (B) A non-reflexive pronoun must not have an antecedent within its local clause.
- (1.15) Ancillary definition:
 α is within ϕ 's *local clause* if α and ϕ are dominated by the same set of clausal nodes (S, $\bar{S}$, IP, CP, TP, AgrP ...).

Exercise 1.1

In the following sentences, Φ designates an NP with the index given. For each sentence, determine by intuition what Φ can/must be (there may be more than one option in some cases). Then give the local clause and the antecedent for Φ and demonstrate that the Binding Conditions in (1.14) are met (example: Φ_3 in [1.16a] must be *himself*, its local clause is the matrix S/IP, and its antecedent is *Peter*, which is, correctly, in the same local clause):

- (1.16) (a) Peter₃ watches Φ_3 in the mirror.
 (b) Masha₅ believes that the swamp elks admire Φ_5 .
 (c) Masha₅ believes that [the swamp elks]₁₆ admire Φ_{16} .
 (d) Masha₅ introduced Φ_5 to the swamp elks.
 (e) Hermann₈ tried to be nice, and Gallia quite liked Φ_8 . Now Φ_8 and Gallia go out to see a mud wrestling show.
 (f) Masha₅ mentioned a swamp elk that was important to Φ_5 .
 (g) Φ_1 's manager takes care of Cecilia₁'s business.
 (h) Φ_1 takes care of Cecilia₁'s business.

1.2.2 Binding and binder

Before going on, we need to refine our previous treatment in one small but significant way. To see why, consider (1.17):

- (1.17) (a) Carlotta₁₁'s dog accompanies her_{11/6} to kindergarten.
 (b) *Carlotta₁₁'s dog accompanies herself_{11/6} to kindergarten.

The judgments in (1.17) are the reverse of what the Binding Conditions lead us to expect: *Carlotta* is clearly in the same local clause as *her/herself*, yet we

have to choose a non-reflexive pronoun to express coreference. This is in marked contrast to our earlier example (1.10), repeated here, which led to the formulation of the Binding Conditions above:

- (1.18) Peter₃ watches himself₃/*him₃ in the mirror.

One difference is that *Peter* and *himself* in (1.18) are *clausemates*, whereas *Carlotta* and *her(self)* in (1.17) are not – *Carlotta* is the possessor to the subject, but only the subject and *her(self)* are clausemates. We can flesh out the notion ‘clausemate’ in various ways, e.g. as ‘be arguments to the same predicate’ (here: *watch*), or ‘be immediate constituents of the same clause,’ with subtly different results, as we will discuss immediately in sections 1.2.4 and 1.3.

Postponing a precise definition of clausemate, let us say that only an antecedent which is a clausemate to an NP can be a *binder* for that NP:

- (1.19) Binding (preliminary): NP₁ binds NP₂ if and only if (iff)
- (a) NP₁ and NP₂ are coindexed
 - (b) NP₁ precedes NP₂
 - (c) NP₁ and NP₂ are clausemates.
- Then NP₁ is the *binder* of NP₂, and NP₂ is *bound* (by NP₁).

(1.19a) and (1.19b) are the same as in the definition of antecedent in (1.3) above, but clause (1.19c) is added. A binder, then, is simply an antecedent that is a clausemate of the bindee. We now replace the notion of ‘have an antecedent’ with the notion of ‘be bound’ in the Binding Conditions:

- (1.20) Binding Conditions (still preliminary):
- (A) A reflexive pronoun must have a binder within its local clause.
 - (B) A non-reflexive pronoun must not have a binder within its local clause.

In (1.18), repeated in (1.21a) below, *Peter* qualifies as a binder with respect to the pronoun in the object position of *watch* – it is coindexed with it, precedes it, and, being the subject of *watch*, is a clausemate. Hence Binding Condition A licenses a reflexive in object position, and Binding Condition B prohibits a non-reflexive. All’s well:

- (1.21) (a) Peter₃ watches himself₃/*him₃ in the mirror.
 (b) Carlotta₁₁’s dog accompanies her₁₁/*herself₁₁ to kindergarten.

In the formerly problematic example (1.17), repeated in (1.21b) above, *Carlotta* is not a binder to the pronoun in the object position of *accompany* (though it is an antecedent); it is coindexed with it, and precedes it, but, being a modifier to *dog* rather than an argument to *accompany*, it fails on the clausemate condition in the definition of binder (1.19c). Binding Condition A thus prohibits a reflexive, and Binding Condition B allows a non-reflexive.

1.2.3 Full NPs

Turning now to full NPs, we observed that they cannot occur with a sentence internal antecedent at all, regardless of whether the antecedent occurs within the same local clause or not. The relevant data are repeated here:

- (1.22) (a) That it rains bothers Peter. (no antecedent)
 (b) *Carla₄/she₄ thinks that I hate Carla₄. (non-local antecedent)
 (c) *Peter₃/he₄ watches Peter₃ in the mirror. (local antecedent)

The question that comes up is whether full NPs are allergic to antecedents, or just binders. To decide that question we have to look again at a case in which an NP antecedes a full NP without actually binding it, for example (1.23):

- (1.23) (a) Her₁₁ dog accompanies Carlotta₁₁ to kindergarten.
 (b) ?Carlotta's₁₁ dog accompanies Carlotta₁₁ to kindergarten.
 (c) Carlotta's₁₁ dog accompanies the little darling₁₁ to kindergarten.

The pronoun in (1.23a) antecedes the full NP with no loss in acceptability. And even another full NP can, as in (1.23b), which is slightly degraded due to the repetition of the name, but head and shoulders above (1.22b); and (1.23c), which features an *epithet*, i.e. a definite NP which is coreferential with, though different in descriptive content from, its antecedent, is impeccable.

We conclude that, just as in the principles governing the coreference options of pronouns, the principle responsible for full NPs must make reference to the notion of binding, rather than antecedence:

- (1.24) Binding Condition C: A full NP must not be bound.

I should like to point out here that the judgments in (1.23), while widely accepted, are not uncontroversial. Generally, name–name cases (*Peter*₃ ... *Peter*₃) seem more acceptable than pronoun–name cases (*he*₃ ... *Peter*₃) and for many speakers approach the degree of acceptability found in examples like (1.23b) (cf. e.g. Bach and Partee [1980], note 11; Evans [1980]:356 a.o.). This can be seen as a phenomenon outside of grammar (after all, in the double name cases, the coreferential reading is the only way to interpret the sentence at all, while in the pronoun–name cases, there is a host of grammatical non-coreferent readings) or as a fact about BT proper, suggesting that Binding Condition C should only ban full NPs from being bound by a *pronoun* Bach and Partee [1980]; Keenan [1974]; for further discussion see also Bresnan [2000], Lasnik [1986], as well as chapter 6. We will, for the time being, assume these cases to be unequivocally bad.

1.2.4 C-command

Before closing, we need to generalize the notion of binding slightly. As it stands, Binding Condition C does not exclude (1.22b), repeated here:

- (1.25) *Carla₄/she₄ thinks that I hate Carla₄.

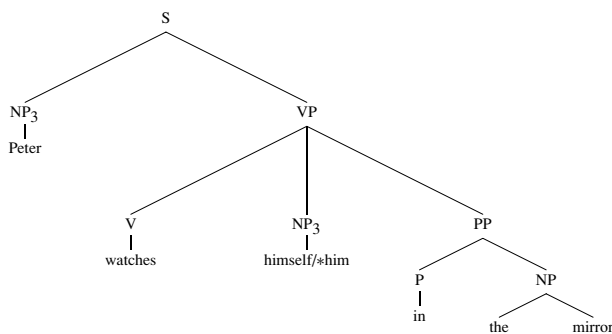
The reason is that (the first occurrence of) *Carla/she* in (1.25) doesn't *bind* the second in the technical sense defined in (1.19), because they are not clausemates:

they are not immediate constituents of the same clause, nor are they arguments to the same verb (*think* versus *hate*). We therefore replace the notion of clausemate by a more general, asymmetric, notion, that of *c(onstituent)-command* [Reinhart, 1976]:

- (1.26) Node A *c-commands* node B in a phrase marker iff
- (a) neither dominates the other, and
 - (b) every (branching) node that dominates A also dominates B³
- (1.27) Binding (revised, still preliminary): NP₁ binds NP₂ iff
- (a) NP₁ and NP₂ are coindexed
 - (b) NP₁ precedes NP₂
 - (c) NP₁ *c-commands* NP₂
- Then NP₁ is the *binder* of NP₂, and NP₂ is *bound* (by NP₁).

Let us first verify how these new definitions subsume the old ones. Take (1.21a), repeated here; a phrase structure tree for this sentence will have the essential constituency shown in (1.28):

- (1.28) Peter₃ watches himself₃/*him₃ in the mirror.

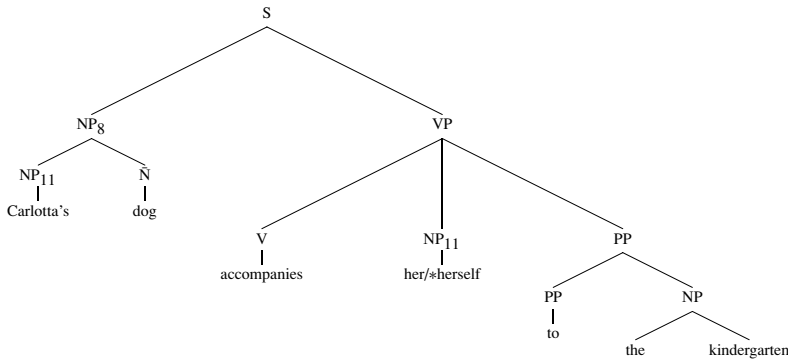


The only (branching) node dominating $[_{NP}Peter]_3$ is S, which means that $[_{NP}Peter]_3$ *c-commands* VP and everything dominated by VP, including $[_{NP}himself/him]_3$. Thus $[_{NP}Peter]_3$ is a binder for $[_{NP}himself/him]_3$, and, given that it is in the same local clause, it is correctly predicted that the latter has to be a reflexive, rather than a full NP or a non-reflexive pronoun.

Contrast this with (1.21b) repeated here along with a simple tree diagram:

- (1.29) Carlotta₁₁'s dog accompanies her₁₁/*herself₁₁ to kindergarten.

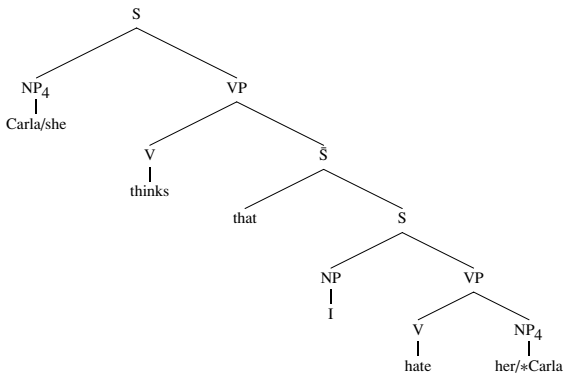
³ Definitions in the literature usually include the qualification 'branching', even though, as Barker and Pullum [1990] and Pullum [1986] note, this is rarely argued for, nor required, by the data in any obvious way. The cases discussed in this book provide no exceptions to that; indeed the notion of semantic binding to be introduced in chapter 4 directly embodies Pullum's stricter and arguably more natural notion of IDV-command, according to which a constituent's *c-command* domain simply consists of its sister constituent(s).



Here, NP₁₁, *Carlotta*, does not c-command VP or anything therein: nodes dominating NP₁₁ are NP₈ and S, which means that NP₁₁ merely c-commands the $\bar{N}$ *dog*; VP, and the pronominal NP₁₁ within it, though dominated by S, are not dominated by NP₈, which means they are not dominated by *every* branching node dominating NP₁₁, *Carlotta*, as is required for binding due to (1.26b). Accordingly, $[_{NP} \text{her}(\text{self})]_{11}$ is not bound by $[_{NP} \text{Carlotta}]_{11}$ by the new definition of binding, especially (1.27c), so that the Binding Conditions correctly predict a non-reflexive (or a name) in that position.

Crucially, the new definition of binding is ‘downward unlimited’, because an NP that c-commands a node A also c-commands every node dominated by A. This is the key to handling the Binding Condition C cases. Consider again (1.22b), repeated here:

(1.30) *Carla₄/she₄ thinks that I hate Carla₄.



Similar to (1.21a), the matrix subject NP₄, *Carla*, c-commands the matrix VP, and everything dominated by the matrix VP, including the object NP₄. Since the subject NP₄ is also coindexed with the object NP₄ and precedes it, it qualifies as a binder. Binding Condition C then excludes a name as the object NP₄, while Binding Condition B allows a non-reflexive pronoun in that position.

This completes our introduction to the ABC of Binding Theory for English. It should be stressed that the Binding Conditions as stated above are no longer

about the traditional, intuitive concept of antecedence, but about a more abstract concept, binding. Binding Theory, so construed, is then a theory only about a subset of anaphoric relations, excluding non-c-command anaphora, both across and within sentences. This embodies a strong and non-obvious hypothesis, namely that c-command, or some other command notion (more about which is discussed in section 1.3), is of utmost significance for BT, and that, accordingly, the data fall into two broad natural classes – binding versus non-c-command anaphora. We will continue to reflect upon the validity of these hypotheses in the course of this book.

On the other hand, if Binding Conditions are indeed based on the notion of c-command, they can serve as a probe into the phrase structure of a sentence: if an NP blocks the occurrence of a coindexed pronoun or full NP', NP must c-command NP'. Binding Condition C in particular will be useful in this regard, since it applies across clause boundaries. It has been suggested, for example, that the pairs in (1.31) and (1.32) show that object clauses, but not temporal adverbial clauses, are c-commanded by the object, while both are c-commanded by the subject:

- (1.31) (a) *The dog told him₁ [that the horse₁ would fall].
 (b) The dog hit him₁ [while the horse₁ ate lunch].
- (1.32) (a) *She₈'ll talk to me [when Sheila₈ gets back from lunch].
 (b) I'll talk to Sheila₈ [when she₈ gets back from lunch].

While this method can be useful, it should be applied with care, for at least two reasons: first, as pointed out in section 1.2.3 above, the unacceptability of bound full NPs is itself not uncontroversial, and judgments seem to vary between speakers, but also in response to prosodic, stylistic, and discourse-pragmatic factors (see e.g. Carden and Dieterich [1981]; and Gerken and Bever [1986] for experimental results). Second, subordinated clauses are often found in displaced positions (e.g. through topicalization or extraposition), or at least could be for all we know, so that our conclusions from such examples rely in turn on our conclusions about the interaction of Binding Conditions with displacement (more on which in chapter 12). We will suggest that the phenomenon of *semantic binding*, to be introduced in chapter 4, may provide a more reliable diagnostic for c-command. Since we are presently concerned with demarcating the conditions on binding themselves (rather than presupposing them to figure out constituency), we will for the most part ignore constructions whose constituent structure is itself subject to debate.

1.2.5 Taking stock

It will be useful to separate several parts or components of the theory, as these will be subject to criticism, revision, or modification later, independent of each other:

- The **classification** of NPs according to their coreference and binding options. Here: three classes, reflexives and reciprocals ('anaphors'), non-reflexive pronouns ('pronominals'), non-pronominal or full NPs ('r-expressions').
- The identification of one or more **domain(s)** within which binding requirements apply. Here: the minimal clause.
- The formulation of a proper notion of **command** or *accessibility* as prerequisite for, and source of, asymmetry in binding. Here: precedence and c-command.

The general format of a Binding Condition can then be schematized as in (1.33):

- (1.33) An NP of **class** must (**not**) be coindexed with a **commanding** NP within its **domain**.

Developing a general theory of binding is to formulate conditions of the general form in (1.33) for several languages, and in the process, to determine which exact values for the variables **class**, **command**, and **domain** are empirically most adequate, within a language, and cross-linguistically, and which are systematically irrelevant. In this book, we will be concerned with the notion of domain in chapter 3, the issue of classification in chapters 3 and 11, and the concept of command in section 1.3 of this chapter, as well as in chapter 12. Hopefully, the tripartition of the ingredients to the BT will prove useful in keeping track of the discussion.⁴

We have set up our system in such a way that each class of NPs may have (a) positive and/or negative Binding Condition(s) associated with it. Complementary distribution between two classes results if the negative binding domain for one element happens to be the same as the positive binding domain for the other. There is nothing in the formal system that accounts for the intuition one might have that one class of NPs is used *because* of the unavailability of the other, and vice versa. Intuitions of this kind have motivated so-called *blocking* approaches to Binding Theory, in which one form is used if (and perhaps only if) the other is excluded; examples of such approaches include Dowty (1980); Farmer and Harnish (1987); Huang (2000); and Levinson (1987, 1991, 2000); in light of the fact that non-complementary distribution between different pronoun classes is common across languages, including English (see chapters 3 and 11), we will not review these approaches further in this book; but see Burzio (1996, 1998); Kiparsky (2002) for refined, hybrid blocking approaches.

Exercise 1.2

A naïve approach to binding would be that the use of a pronoun is necessary to avoid repetition of full NPs, especially within a single sentence.

⁴ Most authors appear to assume, implicitly, that the notion of command, once defined properly, is invariant across all Binding Conditions across all languages (but see the remarks at the end of section 1.3); if so, the variable **command** shouldn't be treated on a par with the others; we will leave this issue open in this book.

Use the distinction between antecedence and binding established above to make an argument against this view. Construct and provide crucial examples.

Exercise 1.3

Both sentences in (1.34) below could be used to illustrate that reflexives in English cannot be bound across a clause boundary. Explain how each does it. Can you think of reasons to prefer one mode of presentation over the other?

- (1.34) (a) John₁ thinks that Bill₂ likes himself_{2/*1}.
 (b) *John thinks that I like himself.

Exercise 1.4

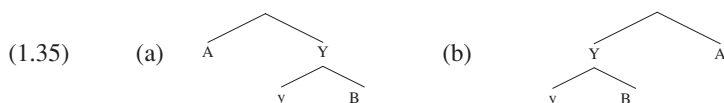
Consider the English possessive form *his/her/its own* (pretend that it is a single form). What requirements, if any, does it impose on its binder or antecedent? Formulate an appropriate Binding Condition for it, and adduce (acceptable and unacceptable) examples to support your proposal.

1.3 Command and precedence^⑥

As discussed in 1.2.5 above, the notion of *command* is one key ingredient in the formulation of Binding Conditions. We have used c-command plus precedence in our definition of binding above, hence as our relevant notion of **command** in (1.33). In this section we will critically reexamine these two notions, suggesting that they should perhaps be replaced with an altogether different command notion.

1.3.1 Against precedence

In our final definition of binding in (1.19) we have used two relations that give us an asymmetrical ordering among the NPs in a sentence, c-command and precedence. Is this necessary, or even tolerable? Consider the abstract phrase markers in (1.35):



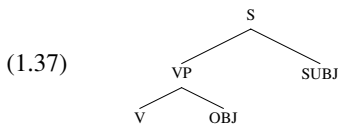
In (1.35a), the standard kind of case we have been looking at in English, A could bind B because A c-commands and precedes B; and B could not bind A, because it neither c-commands nor precedes A. In other words, according to our present definition of binding, (1.27), there are two different reasons why A can bind B, but not the other way around; the account is thus somewhat redundant, though perhaps harmlessly so.

In (1.35b), on the other hand, A c-commands B asymmetrically, but B precedes A. If both factors are relevant, binding between A and B should be *ineffable*, i.e. impossible either way around, and regardless of the morphological shape of either NP. While English lacks clear instantiations of this configuration, other languages arguably provide them. Reinhart (1983a: 47) provides the following data from Malagasy (attributed to E. Keenan, p.c.), an Austronesian language spoken in Madagascar (for convenience, I have set the pronouns in boldface and the antecedents in italics):

- (1.36) (a) namono **azy** ny anadahin- d- *Rakoto* (Malagasy)
 hit/killed him the sister- of- Rakoto
 ‘Rakoto’s sister killed him.’
 (b) *namono ny anadahin- d- *Rakoto* **izy**
 hit/killed the sister- of- Rakoto he
 ‘He killed Rakoto’s sister.’

Malagasy is a VOS language, but, as the translations make clear, behaves rather like English with respect to the BT. In particular, (1.36b) appears to be a Binding Condition C violation, with the subject pronoun *izy*, ‘he’, illicitly binding the full NP *Rakoto* within the object NP. No such effect is found in (1.36a), in which the full NP is the subject, and the pronoun is in the object.

The data then suggest that the subject can bind the object, but not vice versa. Given that the object precedes the subject, this pattern straightforwardly prohibits a treatment in terms of precedence. C-command alone, however, would seem to provide the correct asymmetry, provided we assume that the basic clause structure of Malagasy is essentially as in (1.37) (cf. [1.35b] above):



We thus conclude that the inclusion of precedence in the definition of binding, while perhaps merely redundant for English, is actually harmful if we want to apply the notion cross-linguistically, and should be dropped.⁵

1.3.2 Limitations of C-command

Unfortunately, c-command is not unproblematic either. Languages that display flexible constituent ordering abound with examples in which a bound element precedes and c-commands its binder. Japanese, Korean, and German provide three examples. They are strictly or mostly left-branching languages with

⁵ I found exactly one case in the literature in which precedence does seem to play a role, namely Samoan, as discussed in Chapin (1970); and Keenan and Stabler (1995). Whether this requires inclusion of precedence in the BT for Samoan, or can be captured in any other way, will be left as an open question here.

free word order, in all of which we find, for example, (reflexive-like) pronouns preceding coreferring full NPs:⁶

- (1.38) **Zibun** -o *Hanako* -ga utagatte iru (Japanese)
self -ACC *Hanako* -NOM *doubts*
 ‘Hanako doubts herself.’
- (1.39) **Caki casin** -eke *Kim* -ŭn silmanghaössta (Korean)
self -DAT *Kim* -TOP *disappointed*
 ‘Kim was disappointed in himself.’
- (1.40) Oft hat **sich der Mann** im Spiegel betrachtet. (German)
often has self the man in the mirror watched
 ‘Often the man watched himself in the mirror.’

These data are problematic for our definition of binding, because, unlike in the case of Malagasy discussed above, there is agreement that none of Japanese, Korean, or German has a constituent structure in which the reflexive objects in these examples are c-commanded by their subject antecedents.

It should be noted, however, that in all these examples, what could be called the ‘logical subject’ binds the ‘logical object’, regardless of order or c-command. In other words, one could conjecture that some more abstract level of representation displays an asymmetric ordering among constituents, at which all of the examples discussed in this section show the relative ordering in (1.41):

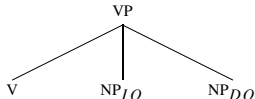
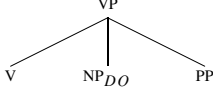
- (1.41) ‘subject’ > ‘indirect object’ > ‘direct object’ > ‘prepositional object’

I have scare-quoted the notions ‘subject’, ‘object’, etc. because these are merely place-holders for whatever theoretical constructs a particular theory regards as relevant here. We will inspect several instantiations of such a hierarchy in turn.

Before that, I want to bring up another set of data on which the c-command condition, even if it doesn’t fail as spectacularly as on those in (1.38–1.40), seems deficient, namely so-called non-configurational or ‘flat’ structures. It is, for example, generally accepted that the objects in double object constructions in English show asymmetries in binding behavior:⁷

- (1.42) (a) I showed John himself (in the mirror).
 (b) *I showed himself John (in the mirror).
- (1.43) (a) I showed Mary to herself.
 (b) *I showed herself to Mary.

Suppose that the structure of the VP in double object constructions is as in (1.44):

- (1.44) (a)  (b) 

⁶ Korean data from Keenan (1988):131.

⁷ Reflexive examples from Barrs and Lasnik (1986):347 and Larson (1988):338.

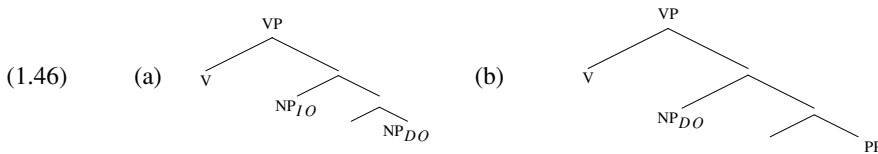
Given this structure, no asymmetries in binding are expected, contrary to fact, since the two objects mutually c-command each other, at least in the double NP case. Similar remarks apply, for example, to nouns with more than one post-nominal attribute:⁸

- (1.45) (Ja čital) stat'ju Tolstoja o sebe. (Russian)
I read article-ACC T.-GEN about himself-LOC
 'I read) an article of Tolstoj about himself.'

Again, the immediately post-nominal phrase can bind the second one, but not vice versa. But, as in the case of verbal double-object constructions, the constituent structure appears to be flat.⁹

Both these examples would be treated correctly if we assumed c-command and precedence as prerequisites for binding, but, in the light of the problems with precedence encountered earlier, it is worth while to look for alternative treatments. A hierarchy along the lines of (1.41) above can provide such a treatment, as it provides a total ordering even among NPs that mutually c-command each other.

To be sure, there have been attempts to reconcile the data in (1.42–1.43) with a c-command approach to binding, notably Larson (1988), which argues that the asymmetries rather show that these structures are more complex than (1.44) and have the essential properties of those in (1.46):



We do not have to elaborate on the details of this analysis (like the identity of the unlabelled nodes in [1.46]). The important thing is that a structure like (1.46) predicts the binding asymmetries as a function of asymmetrical c-command, without reference to anything else.

On the other hand, we saw the need to reconsider the c-command condition anyway in the light of the Japanese/Korean/German cases, and the two alternatives to c-command presented below are intended to capture those and 'flat structure' cases by the same mechanism, namely an independent, non-phrase-structural hierarchy.

This is obviously not the place to rule a final verdict on the (non-)existence of flat structures, but it bears mentioning that the binding facts alone can be treated by a refined command notion, which is presumably required independently, and that structures like in (1.46) are not necessitated by them. We will look at two such notions in what follows.

⁸ Data based on Rappaport (1986):106.

⁹ Note that the simplest binary branching structure would actually be $[[V/N \text{ } XP_1] \text{ } XP_2]$, which would yield the opposite asymmetries in c-command from what we find with the binding data.

1.3.3 Θ -command

Let us start with the perhaps most attractive version of a non-phrase-structural ordering among the NPs, namely the idea that this ordering is essentially semantic in nature, more precisely that it is based on the semantics of the *thematic relations* involved. An early and representative example of this approach is Jackendoff (1972):148, where it is proposed that the relevant asymmetric ordering among elements is derived from the hierarchy of *thematic roles* (or Θ -roles for short) in (1.47):

(1.47) Agent $\succ$ Location, Source, Goal $\succ$ Theme

The fact that *der Mann* can bind *sich* in the German example (1.40) above is then due to the fact that *der Mann* bears the Θ -role ‘Agent,’ which outranks the role of *sich*, ‘Theme’ on the Θ -hierarchy in (1.47). Likewise, *Rakoto* must not be coreferent with *izy* in the ungrammatical Malagasy example in (1.36), due to the fact that *izy* has the ‘Agent’ role in that sentence, which is higher than any other role in the Θ -hierarchy. The initial appearance that binding asymmetries correlate with a hierarchy among phrase-structure positions (as established by c-command) is merely an epiphenomenon of the dependence of phrase structure upon this same thematic hierarchy.

There are two more immediate advantages of Θ -command over c-command. First, as it is defined over thematic roles, it captures certain parallelisms between NPs and PPs. Not only can binding proceed ‘into’ PPs – a fact captured by c-command and Θ -command alike – sometimes it can also proceed ‘out of’ PPs, if the pertinent thematic command obtains:

- (1.48) (a) We talked to John about himself.
 (b) *We talked to himself about John.
 (c) *We talked about John to himself.

These examples (from Wilkins [1988b]:208) show that Goal (as realized by a complement of *to*) Θ -commands Theme (as realized by a complement of *about*) and can therefore bind it, regardless of the fact that *John* doesn’t c-command *himself*, because the first node dominating it is the PP.

Second, Θ -command allows a straightforward account of binding with so-called *psych-verbs*. With these verbs, a phrase-structurally lower argument appears to bind into a higher argument:¹⁰

- (1.49) (a) Stories about herself generally please Mary.
 (b) Each other’s health worried the students.

While obviously unexpected from a phrase-structure point of view, these examples can be captured under the assumption that the role borne by the subject of

¹⁰ Examples from Pesetsky (1987):127.

these verbs, Theme, is lower on the Θ -hierarchy than that of the object, Experiencer.¹¹

Attractive though the Θ -command notion is for the purpose of BT, it also faces some serious problems. First, verbs that allow alternative realizations of their arguments, such as English double-object constructions, show alternations in binding possibilities as well, as discussed at length above, cf. the examples in (1.42) and (1.43) above; (1.50) reiterates the point:¹²

- | | | | |
|--------|-----|------------------------------|--------------------------|
| (1.50) | (a) | I sold the slave himself. | (Goal binds Theme) |
| | (b) | I sold the slave to himself. | (Theme binds Goal) |
| | (c) | *I sold himself the slave. | (Goal cannot bind Theme) |

A Θ -command approach has to grab the bull by the horns and claim that, say, *John* bears a different Θ -role in *I sold the cabbage to John* than in *I sold John the cabbage*, as is done in Wilkins (1988b):208ff., who claims that a Theme, when verb-adjacent, as in (1.50b), is also a *Patient*, which is a higher role than Goal, whereas it is only a Theme if it follows another object, as in (1.50a) and (1.50c), which makes it lower than Goal (a similar strategy is implemented in Jackendoff [1990]'s *action tier*).

This strategy, while technically feasible, highlights a more general weak spot of Θ -based approaches, namely the question of how to determine the role of a given argument in a given structure. For example, according to its proponents, the Theme/Patient distinction *explains* what is behind the dative alternation in English. According to the critics, it merely gives a semantically loaded name to what by all appearances is a purely syntactic distinction, namely that between direct and oblique object (alleged semantic distinctions between the two are indeed subtle, but see Krifka [2004]). According to critics, then, a sufficiently elaborated Θ -command account is just a somewhat misleadingly labelled instantiation of an argument-structure based account, to be discussed in the next subsection.

Before closing this section, let us briefly consider two other areas in which the Θ -command idea has been argued to encounter difficulties. In certain cases of *raising*, reflexivization involves NPs that do not receive their thematic role from the same lexical element. Consider (1.51):¹³

- (1.51) Max strikes himself as qualified for the job.

It would appear that the Θ -role of *himself* (Goal or Experiencer of *strike*) is higher than that of *Max* (Theme of *qualified*) as well as that of the clause *Max has raised* from (which is presumably the Theme of *strike*). Yet, binding can only proceed from the subject to the object position, in violation of the Θ -command condition.

¹¹ The facts are more complicated, as must therefore be the account; see, among many others, Belletti and Rizzi (1988).

¹² From Pollard and Sag (1992:298); in contradistinction to Pollard and Sag, Larson (1988), and others, Jackendoff (1972:157) and Postal (1971:ch. 15) judge both variants of binding in double object constructions as ungrammatical.

¹³ cf. Pollard and Sag (1992):299.

The second argument asserts that morphological processes that change grammatical functions also change binding options. Similar to the case of verbs with alternating argument structures, no obvious change in thematic roles goes along with it:¹⁴

- (1.52) (a) John shaved himself. (Agent binds Theme)
 (b) John was shaved by himself. (Theme binds Agent)
 (c) *We expect himself to be shaved by John (Agent cannot bind Theme)

This argument has to be taken with a grain of salt, though. Many speakers find examples like (1.52b) rather marginal. Cross-linguistically, the diagnosis is actually much less clear, perhaps non-uniform. Toba Batak (Malayo-Polynesian; Sumatra) is a verb-initial language with no morphological case distinctions.¹⁵ Depending on the choice of verbal prefix, *mang* versus *di*, a sentence of the form *V NP₁ NP₂* is interpreted with *NP₁* as Theme (or Patient) and *NP₂* as Agent (*mang*-prefix), or with *NP₁* as Agent and *NP₂* as Theme (*di*-prefix). In other words, *mang*- and *di*- determine different mappings between thematic roles and phrase-structure positions and are, as such, comparable to grammatical function-changing operations like passive in other languages. Strikingly, however, the binding always goes from Agent to Theme, regardless of prefix choice and, accordingly, linear order.¹⁶

- (1.53) (a) Mang -ida si Ria si Torus (a') Di -ida si Torus si Ria
 MANG see the R. the T. DI see the T. the R.
 'Torus saw Ria.'
 (b) * Mang -ida si Torus dirina (b') Di -ida si Torus dirina
 MANG see the Torus self DI see the Torus self
 'Self saw Torus.'
 (c) Mang -ida dirina si Torus (c') * Di -ida dirina si Torus
 MANG see self the Torus DI see self the Torus
 'Torus saw self.'

It appears that neither the change in linear order (and the change in c-command relations presumably accompanying it) nor the change in grammatical function indicated by the verbal prefixes affects the binding options between the two NPs. A good candidate for what remains invariant in all structures are the Θ -relations: *Torus* is the Agent and *Ria* is the Theme. We conclude that grammatical function-changing operations do not provide clear evidence for or against a Θ -command approach to Binding Theory.

¹⁴ Examples from Pollard and Sag (1992):298.

¹⁵ See Keenan (1988) and Schachter (1974); Schachter argues that the basic phrase structure is $[[V\ NP]\ NP]$, which would make it a VOS language, but nothing hinges on that, cf. also Sternefeld (1992).

¹⁶ Data from Keenan (1988):129f. and Keenan and Stabler (1995):4; very similar data are reported for Balinese in Wechsler and Arka (1998):406ff.

1.3.4 Obliqueness-command

The last command relation to be discussed in this chapter is that of *obliqueness command*, or *o-command* for short (sometimes also called a(rgument)-command for reasons to become clear shortly). O-command occupies a middle ground in between c-command and Θ -command in that it claims that there is a non-thematic, syntactic ordering between the NPs in a sentence, but denies that that ordering is directly encoded in the phrase structure as c-command (or any other phrase-structure relation for that matter). I illustrate the idea using an argument-list notation;¹⁷ a verb like *watch*, for example, will have an a(rgument)-list like in (1.54):

(1.54) a-list of *watch*: $\langle \text{NP}, \text{NP}_{acc} \rangle$

The order of elements on the a-list signals *obliqueness*. Thus the first NP on the a-list in (1.54) is less oblique than the second, NP_{acc} . Obliqueness does not necessarily correspond to linear order; if, for example, a language allows different linearizations of arguments, the obliqueness relations encoded on the a-list still remain the same. Thus the a-list associated with *betrachten*, ‘watch,’ in the German example (1.40) above is the same as in (1.54), despite the fact that the accusative NP, the reflexive *sich*, linearly precedes the subject NP (and presumably also c-commands it). Similarly for the Japanese and Korean examples discussed.

Likewise, a flat constituent structure like in (1.44) is perfectly compatible with a total ordering in terms of obliqueness (in fact, by its very nature an a-list is always totally ordered), as, for example, in (1.55):

(1.55) a-lists for ditransitive verbs like *give*:
 (a) $\langle \text{NP}, \text{NP}_{acc}, \text{PP}_{to}[\text{P-OBj}] \rangle$
 (b) $\langle \text{NP}, \text{NP}_{acc}, \text{NP}_{acc} \rangle$

Using argument-lists, we can now define an alternative notion to c-command, *obliqueness-command*, and use that relation to define the Binding Conditions (cf. Pollard and Sag [1992]:287):¹⁸

(1.56) (a) *A locally o-commands B* iff A is less oblique than B on some a-list
 (b) *A (locally) o-binds B* iff A and B are coindexed and A (locally) o-commands B.

¹⁷ This is similar to the SUBCAT or ARG-ST list of *Head Driven Phrase Structure Grammar* as presented in Pollard and Sag (1994); Sag and Wasow (1999) or the a-structure of *Lexical Functional Grammar* (Bresnan, 2000, and the references therein). In current HPSG, the ARG-ST list is split up into separate lists, which between them capture what is traditionally called the valency and subcategorization, partly in recognition of the problems languages like Toba Batak and Balinese, as mentioned on page 18 pose for theories that equate argument structure with grammatical relations (Manning and Sag, 1999; Wechsler and Arka, 1998).

¹⁸ These definitions, as well as that of o-command below, assume that the elements on the a-list are the actual syntactic objects, as they in fact are in HPSG, not just category names. For details, see Pollard and Sag (1994). Otherwise we would have to say that A locally o-commands B (in S) if A' is less oblique on some element's a-list in C than B', where A' and B' are the elements on the a-list corresponding to A and B.

(1.57) Binding Conditions:

(A) A reflexive pronoun must be o-bound within its local domain.

(B) A non-reflexive pronoun must not be o-bound within its local domain.

To define Binding Condition C we must generalize the notion of local o-command to o-command *simpliciter*. The idea is clear enough: if A locally o-commands B, and C is dominated by B, then A should o-command C:

(1.58) O-Command: A o-commands B iff

(a) A locally o-commands B, or

(b) A locally o-commands a C which dominates B¹⁹

(1.59) Binding Condition C: A full NP must not be o-bound.

To give an example, (1.60a) will be ruled out as a Binding Condition C violation, as it should be, given the a-list of *tell* in (1.60b):

(1.60) (a) *The red-haired baroness told him₁ that Casimir₁ is nice.

(b) a-list for *tell*: ⟨ NP, NP_{acc}, $\bar{S}$ ⟩

NP *him* locally o-commands $\bar{S}$, *that Casimir is nice*, which in turn dominates the NP *Casimir*. Therefore, *Casimir* is o-commanded by *him* and, since they are coindexed, also o-bound by it. This binding violates Binding Condition C in (1.59), hence coreference is impossible.

O-command based approaches to Binding Theory are presumably the most successful, since they allow for a certain leeway relative to the thematic relations, and relative to the constituent structure. But then this is also a point of potential criticism, which would insinuate that a-lists can simply be written so as to match the binding facts. Put differently, the ordering of elements on a-lists should ideally have other functions than just encoding binding asymmetries, and there should be a tight correspondence between a-lists and thematic relations, on the one hand, and a-lists and constituent structure, on the other, to account for the sort of facts that motivate Θ - and c-command based approaches.

Arguably, most approaches within the *Principles and Parameters* framework are in essence (equivalent to) o-command approaches. While they assume that the relevant command notion is c-command, Binding Conditions make reference not to the level of surface constituency but to a more abstract level at which c-command and binding go hand in hand. This level, then, encodes in a phrase-structural manner the same ordering that is encoded non-configurally on a-lists under the o-command approach. Not surprisingly, this level is one which, overall, reflects grammatical-function changes such as passive, raising, and dative shift, and ignores or undoes more ‘superficial’ movements such as topicalization, *wh*-movement, and ‘scrambling’ as found in Japanese, Korean, and German (cf. chapter 12). And, as with the o-command approaches discussed above, the more it can be shown that the pertinent level serves other purposes as well,

¹⁹ Cf. Pollard and Sag (1992):300; see also the remarks in Bouma *et al.* (2001):8f., 44.

and is linked to surface constituency and thematic information by a rather constrained inventory of transformations, the more such an approach recommends itself.

With these remarks we leave the issue of command. Of course, none of the above possibilities is exclusive, and there exist approaches in which different command relations coexist (e.g. É. Kiss [1992]), just as there exist approaches on which Binding Conditions must apply at more than one syntactic level (e.g. Belletti and Rizzi [1988]). With the exception of chapter 12, which is exclusively devoted to this issue, we will concentrate on cases in which any one of the command notions yields the same ordering among NPs in what follows, which, at least for English, is easily done. That this is possible reflects the fact that all approaches discussed above converge on their core finding. In particular, they all agree that there *is* an asymmetric and transitive command ordering among the NPs in a given structure which determines their binding potential.

Exercise 1.5

Consider the following data from Albanian, taken from K. Williams (1988). Describe the pattern of reflexive binding in terms of grammatical function (subject, accusative object, dative object) first. Then try to put it in terms of one of the command hierarchies above. Argue why you chose the hierarchy you did.

- (1.61) (a) Artisti₁ ia tregoi veten_{1/2} Dritës₂.
artist CLITIC show self-ACC Drita-DAT
 (b) Artisti₁ ia tregoi Dritës₂ veten_{1/2}.
artist CLITIC show Drita-DAT self-ACC
 (c) Artisti₁ ia tregoi Dritën₂ vetes_{1/* 2}.
artist CLITIC show Drita-ACC self-DAT
 'The artist showed Drita herself.'

Exercise 1.6

Consider the following data from Hungarian, from É. Kiss (1992):247. What can you say about the relevant command notion in Hungarian?

- (1.62) (a) A lányokat megmutattam egymásnak. (Hungarian)
the girls-ACC showed-I each other-DAT
 'I showed the girls to each other.'
 (b) A lányoknak megmutattam egymást.
the girls-DAT showed-I each other-ACC
 'I showed each other to the girls.'

1.4 Reflexive verbs and reflexive phraseologies ^⑤

Let me close by adding some observations about reflexive pronouns and their functions. Compare the following sentences:

- (1.63) (a) Spencer behaved himself.
 (b) Spencer despised himself.

Himself in (1.63b) functions as a semantic argument (to *despise*), but *himself* in (1.63a) does not. We can conclude this from the syntax, because we can't replace the reflexive in (1.63a) with a full NP, say *Carmen*, the way we can in (1.63b);²⁰ similarly, we have a semantic intuition that despising is a two-place relation, whereas behaving is a one-place property (though it turns out to be hard to substantiate this intuition further). Verbs like *behave* (*oneself*) are called *inherently reflexive verbs*; they are semantically intransitive, but syntactically transitive, and show a – presumably uninterpreted – reflexive as the semantically 'inert' argument. Examples of this class are few in English (including *disgrace/betake/vaunt oneself*), but are found in great numbers in other languages, among them the rest of Germanic and Romance.

It is open to speculation why languages have inherently reflexive verbs (instead of proper intransitives), though it seems no coincidence that such verbs are often historically related to transitives. In some cases, we can see this relationship even synchronically, as is the case for the causative/inchoative alternation in German, as found in *etwas schließen* – *sich schließen*, *etwas aufhellen* – *sich aufhellen* ('to close [something]', 'to lighten [something] up').²¹

The opposite of verbs like *behave*, if you will, are verbs like *shave* and *wash*, which can be syntactically intransitive but are clearly semantically transitive (*Quinn shaves* and *Quinn shaves himself* both express that Quinn is the Agent and the Theme of the shaving). *Shave*-type verbs, in English and cross-linguistically, very often describe typically self-directed actions such as acts of grooming (cf. English *wash*, *shave*); typically other-directed actions such as seeing, beating, or killing are unlikely candidates to be expressed by (optionally) intransitive verbs, and require a transitive construction with a reflexive pronoun when used to describe a self-directed event.

A strikingly parallel distinction is found in many languages that have complex as well as simple reflexive pronouns (like the continental Germanic languages discussed in chapter 3, section 3.5.1): typically self-directed verbs occur with a simple reflexive, while typically other-directed verbs require the complex form. Descriptively, we thus find the following scale of transitive constructions:²²

²⁰ Other diagnostics for semantically inert reflexives include their inability to be stressed or fronted, to be coordinated, or right-node-raised in conjunctions with non-inherent reflexive verbs.

²¹ Note that the reflexive alternants do not entail any implicit agent, as, say, passives do; *sich aufhellen* is thus the proper translation of *to lighten up* in *The sky lightened up*. Another formally similar construction is the middle construction, which is realized without an object in English, but with a simple reflexive object in many other languages, but I do not want to speculate about the question whether middle constructions are semantically transitive or not here (see Kemmer [1993]; Steinbach [2002], and the references therein).

²² Hellan (1988) argues that self-directed verbs are really ambiguous between an inherently reflexive and a true transitive variant, but see Kiparsky (2002): section 3.8 for convincing arguments against this analysis.

- (1.64) (a) semantically intransitive (inherently reflexive): *behave oneself*, Dutch *zich gedragen* ('behave'); German *sich schließen* ('to close')
- (b) typically self-directed verbs: *shave (oneself)*, *hide (oneself)*, Swedish *tvätta sig* ('shave'), *gömma sig* ('hide')
- (c) typically other-directed verbs: *hate oneself*, *prefer oneself*, Swedish *hata sig själv* ('hate oneself'), *föreda sig själv* ('prefer oneself')

At least historically, the choice of forms in these constructions is likely to be related to the fact that complex reflexives like German *sich selbst* or Swedish *sig själv* are composed of a simple reflexive pronoun and the emphatic marker akin to *self* (the English situation is more complicated, but note that modern English uses the form *himself*, *herself*, etc. as an emphatic marker, as in *The director himself opposed the plan.*; see Eckardt (2002) for a formal semantic analysis). The idea is that hating or preferring oneself (rather than someone else) is the unusual case and deserves special emphasis, but shaving or washing oneself is the norm and doesn't require emphasis (see Haiman [1985, 1995]; König and Siemund [2000]; König and Vezzosi [2002] for cross-linguistic data; note that complex reflexives with verbs of grooming *are* acceptable if, for example, shaving oneself is juxtaposed with shaving someone else first, i.e. if there is contrastive emphasis).

Unsurprisingly, semantically inert arguments, as found with inherently reflexive verbs, cannot bear emphasis (Dutch speakers, for example, find **Hij gedraagt zichzelf*, rather than *zich*, as bizarre as English speakers do **He behaved Carmen*). The fact that a language like English, which lacks a simple reflexive, has extremely few reflexive verbs, and never uses a reflexive in inchoative constructions,²³ may well be related to the emphatic heritage of the reflexive form. In the same vein, note that what are arguably semantically inert NPs (note the oddity of full NPs) in positions other than direct objects occur more frequently in English, in the form of non-reflexive pronouns:

- (1.65) (a) Mary brought her lunch with *her/*herself/*Colin*.
- (b) Bertrand has many friends around *him/*himself/*Cindy*.
- (c) Francesca lost *her/*her own/*Ibrahim's* way.

This suggests that Binding Conditions are relevant for the choice of semantically inert NP forms as well: non-reflexive pronouns cannot occur if they are locally syntactically bound (regardless of whether they are semantically interpreted at all); in addition, (complex) reflexives must not be (or at least abhor being) semantically inert. For more discussion, the reader is referred to Kiparsky (2002) and the references therein.

²³ Or middles, cf. n. 21.

Exercise 1.7

Try to use reflexives, non-reflexives, and full NPs in the blank positions in the examples below. For each, decide whether that position is semantically inert or active; argue carefully!

- (1.66)
- (a) Mr Mitchum wouldn't perjure _____.
 - (b) The tactic is to draw speculators' fangs by making them commit _____ to buying put options.
 - (c) Carmen has a Spanish look about _____.
 - (d) The general looked around _____.

2 Interpreting indexed structures

In the previous chapter we introduced indexing conventions that expressed our intuitions about the (im)possibility of coreference between two NPs in a given structure. In this chapter I am going to provide a semantic interpretation that cashes out this indexing in semantic terms. It is against this formal, precise interpretation of syntactic indexing that we are going to check the predictions of our theory and its possible modifications later on.

2.1 Basics of interpretation

The kind of semantic interpretation we assume here is a truth-conditional one. Linguistic expressions are associated with non-linguistic entities, i.e. things in the world (e.g. my left thumb and the Cologne Cathedral), and set-theoretic constructs made out of these (e.g. the set of all my fingers, or the set of all sets that contain that set). The task of the semanticist is to devise basic meanings for the words of the language and systematic ways of combining them so as to arrive at intuitively correct truth conditions for entire sentences.

Since this book is not an introduction to semantics, I will keep the technical apparatus to a minimum. Also, common semantic concepts and techniques will be introduced only very briefly. The formalism introduced and used starting with section 2.3 is for the most part compatible with that laid out in great detail in Heim and Kratzer (1998). Readers with a desire for more background are referred to this excellent introduction (other introductory textbooks to formal semantics will for the most part provide the same information, though technical details may differ).

We assume an *interpretation function* $\llbracket \cdot \rrbracket$, which assigns an appropriate meaning or *denotation* to every syntactic object. We say that α *denotes* $\llbracket \alpha \rrbracket$. What are meanings? I just give some examples:

- (2.1)
- (a) $\llbracket \text{Caroline} \rrbracket$ = the real Caroline in flesh and blood
 - (b) $\llbracket \text{Caroline and Fritz} \rrbracket$ = the plurality consisting of the real Caroline and the real Fritz (we'll come back to what a plurality is in chapter 9)
 - (c) $\llbracket \text{smile} \rrbracket$ = the set of all those who smiled
 - (d) $\llbracket \text{Caroline smiled} \rrbracket$ = 1 iff ('if and only if') Caroline smiled, 0 otherwise

$\llbracket \cdot \rrbracket$ must consist of two blocks: a *lexicon*, from which we get meanings for words/terminal nodes such as (2.1a) and (2.1c); and a set of *composition rules* that tell us how to derive the meaning of a non-terminal node from the meaning(s) of its daughter(s), for example:

$$(2.2) \quad \llbracket \text{NP VP} \rrbracket = 1 \text{ iff } \llbracket \text{NP} \rrbracket \text{ is in (an element of) } \llbracket \text{VP} \rrbracket, 0 \text{ otherwise}$$

The numbers 0 and 1 used in (2.1d) and (2.2) are *truth values*. We can think of $\llbracket S \rrbracket = 1$ as saying ‘S is true’, and $\llbracket S \rrbracket = 0$ as ‘S is false’. A statement of the form in (2.1d) ($\llbracket S \rrbracket = 1$ iff ...), which is an instantiation of (2.2) with NP = *Caroline* and VP = *smiled*, is a description of a sentence’s *truth conditions*. Also, we speak of the meta-language statement following the *iff* (e.g. ‘Caroline smiled’ in [2.1d]) as the *proposition* expressed by the sentence.

What is most important for our purposes is that the denotation of a singular referring NP – its contribution to the meaning of a sentence – is a thing out there in the world, a person, thing, or place, or as is common to say in semantic jargon, an *individual* or *entity* (as remarked, we will postpone discussion of the denotation of non-referential NPs until chapter 4, and of plural NPs until chapter 9). We will say that an NP *refers* to that individual.

Exercise 2.1

Give denotations for the words *Carl* and *sleeps*. Then derive the truth conditions for *Carl sleeps* along the lines of (2.1) above.

2.2 Enter indexing

2.2.1 Reference and assignments

In section 2.1 we saw that the interpretation function $\llbracket \cdot \rrbracket$ determines *the* proposition expressed by a sentence, e.g. that *Caroline smiles* for the sentence *Caroline smiles*. For a sentence with a pronoun in it, say *She smiles*, however, we can’t assign *the* proposition it expresses, because it can express as many different propositions as there are individuals that can be referred to using the pronoun *she* (e.g. that *Caroline* smiled, that *the queen* smiled, that *the Titanic* smiled...). The best we can do is to say which propositions such a sentence *can* express (and thereby indirectly say which propositions it *cannot* express).

We therefore augment the interpretation function $\llbracket \cdot \rrbracket$ with an *assignment function*, or *assignment*, for short, written as *g*. The assignment determines which individual a pronoun refers to.

An assignment is a function from the set of natural numbers to individuals. Since indices are natural numbers, an assignment assigns an individual to each index. You can think of an assignment *g* as a sequence of individuals; $g(n)$ then is the *n*-th element in that sequence. (2.3) through (2.5) are three different

assignments:

(2.3)	1	2	3	4	5	6	7	...
	↓	↓	↓	↓	↓	↓	↓	
	Bo	Sven	Kim	Ana	Kurt	Eszter	Tanya	

(2.4)	1	2	3	4	5	6	7	...
	↓	↓	↓	↓	↓	↓	↓	
	Bo	Kim	Sue	Marge	Kurt	Eszter	Tanya	

(2.5)	1	2	3	4	5	6	7	...
	↓	↓	↓	↓	↓	↓	↓	
	Ana	Kim	Marge	Sven	Kurt	Kim	Tanya	

As a first stab, we can now interpret third person singular pronouns as the n -th member of the assignment g , where n is the index on the pronoun. Thus $\llbracket \text{she}_3 \rrbracket^g$ denotes, or refers to, $g(3)$, which is an individual, just like $\llbracket \text{Caroline} \rrbracket^g$. Relative to the assignments in (2.3)–(2.5), it refers to Kim, Sue, and Marge, respectively. The sentence *She₃ smiled*, then, can express the propositions that Kim smiled, that Sue smiled, that Marge smiled, etc.

A different way to think of an assignment is as providing the (relevant aspects of the) *context*. *She₃ smiled* expresses that Sue smiled in the context/assignment (2.4), but that Marge smiled in the context/assignment (2.5). This view assumes that the assignment somehow reflects properties of a context, for example that Ana, Kim, Marge, etc. have been introduced in context (2.5), perhaps that they have different degrees of salience, etc. In this book we will usually speak of assignments as contexts, though we will have little to say about just how the assignment comes to reflect an actual context (cf. section 2.4.1). In a narrower technical sense, assignments are simply a device to arrive at the set of propositions a sentence can express (cf. section 2.3.2).

A full representation of the context will include more than just an assignment. It has to include, for example, the speaker s and the utterance situation u . That way, we can define lexical meanings such as:

- (2.6)
- (a) $\llbracket \text{I/me/myself/my} \rrbracket^{g,s,u} = s$
 - (b) $\llbracket \text{you/yourself/your} \rrbracket^{g,s,u} = \text{the single person } s \text{ addresses in } u$
 - (c) $\llbracket \text{you}_{pl}/\text{you guys} \rrbracket^{g,s,u} = \text{the plurality of people } s \text{ addresses in } u$

We will henceforth write $\llbracket \alpha \rrbracket^{g,s,u}$ to mean ‘the denotation of α relative to assignment g , speaker s , and utterance situation u ’. $\llbracket \rrbracket$ in its unadorned form (i.e. without g etc.) doesn’t play a role anymore; we will, however, omit s and u where these are of no concern, using $\llbracket \rrbracket^g$.

The use of an assignment function to interpret pronouns is borrowed from predicate logic, where assignments are used to assign values to *variables*. It is therefore common to speak of pronouns functioning as variables (or being interpreted as variables – though strictly speaking this should be ‘interpreted like

variables’).¹ Elements that do not depend on the assignment for their meaning – which in natural language include verbs, prepositions, articles, common nouns, etc. – are accordingly called *constants*.

2.2.2 Binding Theory

We begin by making explicit some of the assumptions from the previous chapter, e.g. that (referring) NPs bear indices:

- (2.7) NP Indexing (preliminary):
Every name, pronoun, and definite NP bears an index. Nothing else bears an index.²

How do we interpret this indexing? Essentially, as said above, we want an NP with index n to refer to $g(n)$ under a given assignment g . NPs, however, have some lexical content which is independent of the particular context. Even pronouns do: the pronoun *she* for example can only refer to a singular female individual. How can we capture this? By representing those context-independent meaning parts as *presuppositions*:³

- (2.8) (a) $\llbracket he_1 \rrbracket^g = g(1)$ if $g(1)$ is male, undefined otherwise
(b) $\llbracket her_5 \rrbracket^g = g(5)$ if $g(5)$ is female, undefined otherwise

Relative to an assignment g , sentence S (with a pronoun) will have a denotation only if g happens to meet the presuppositions of the pronoun (e.g. if $g(5)$ is in fact female); otherwise, $\llbracket S \rrbracket^g$ is undefined. If $\llbracket S \rrbracket^g$ is defined, we get one of the meanings S can have.

That the gender (and number) information is a presupposition rather than part of the meaning proper (e.g. something along the lines of ‘ $\llbracket he_1 \rrbracket^g = g(1)$ and $g(1)$ is male’) can be seen by looking at a sentence like (2.9):

- (2.9) Alfred₅ thinks that she₅ missed the bus.

If the ‘is female’ bit were part of the literal meaning of *she*, (2.9) should mean something like ‘Alfred thinks that he is female and that he missed the bus’, given that *Alfred* and *she* are coindexed. But, obviously, there is no such reading. On the

¹ Note that this semantic notion of ‘variable’ does not coincide with the notion of variable as used in the Government and Binding framework (Chomsky, 1981) where the term variable is used to mean ‘trace of *wh*-movement’.

² In transformationalist terms, we assume that an NP starts out with an index (rather than being indexed in the course of the derivation), although nothing essential hinges on this assumption. This means that indexing takes place at d-structure (or at Merge, or whatever fashion calls it).

³ Additional complications arise in languages such as Italian and German which have *grammatical gender*. For example, the word for moon is feminine in Italian (*la luna*) and masculine in German (*der Mond*). Consequentially, Italian speakers refer to the moon with a feminine pronoun, Germans with a masculine one. The presupposition of the pronoun therefore cannot make reference to the sex of the referent but to the gender of the expression used to designate it. More complications arise in cases where different words of different gender can be used to refer to the same thing, e.g. *der Zug* (masc.) vs. *die Bahn* (fem.) for ‘train’ in German. We will not attempt to represent grammatical gender systems in this book.

As for proper names and definite NPs, we assume, too, that it is the index which determines the reference, whereas the lexical content of the NP adds an (identity) presupposition. This is achieved by the schema (2.10), which, as an illustration, is instantiated in (2.10a) and (2.10b):

- Note that rule (2.10) applies to first and second person pronouns, too, so that applying it to, say, a first person pronoun (cf. [2.6a] above) we get (2.11):

- It is worth noting that, as far as interpretation goes, we wouldn't have to index names, first and second person pronouns, and full NPs for their referent is fixed by their inherent meaning (plus the contextual parameters s and u ; see section 2.2.1) anyway. That is, as mentioned above, we assume that – at least in a given context/utterance situation – an NP like *Peter Stuyvesandt, the founder of New York* and even *the waiter* have a unique referent, regardless of their indexing. The need to index full NPs comes from our desire to implement BT. If we were to assign their referents without looking at the index, how could we formalize facts about coreference by indexing? (A possible answer to this question will be discussed in chapter 6.)

(2.12) In her₆ office, Sarah₆ introduced him₂ to the director₈.
 ↓ ↓ ↓ ↓
 Sarah Sarah Steve Caroline
 NOT: Tatjana Sarah Steve Caroline

(2.13) In her₃ office, Sarah₆ introduced him₂ to the director₈.
 ↓ ↓ ↓ ↓
 Tatjana Sarah Steve Caroline

There is a loophole, though, that we haven't explicitly closed yet. Could a sentence like (2.13), whose indexing meets all Binding Conditions, get an

interpretation as in (2.14), where *him*₂ and *the director*₈ corefer, despite being contra-indexed?

- (2.14) In her₃ office, Sarah₆ introduced him₂ to the director₈.
 ↓ ↓ ↓ ↓
 Tatjana Sarah Steve Steve

For (2.14) to work, g must be an assignment in which the same individual – Steve – occurs twice. Intuitively this is unwanted (and it also clashes with the idea that an assignment is a sequence of individuals, for the same physical individual can occupy at most one place in a sequence, it would seem). Formally, however, an assignment is just a function from numbers to individuals, and nothing is wrong with a function that assigns the same value (individual) to two different arguments (numbers) (in fact, [2.5] above was just such an assignment).

Our intuition when doing the indexing, however, has been that non-coindexing means non-coreference (not just ‘possible non-coreference’). That is, we are interested only in a subclass of assignments, namely those which don’t assign the same individual to two different numbers. In other words, we want to exclude the possibility that two NPs that are not coindexed accidentally refer to the same individual. This is done through (2.15):

- (2.15) Prohibition against Accidental Coreference (PACO):
 $\llbracket S \rrbracket^g$ is a possible interpretation of sentence S only if $g(n) \neq g(m)$ if $m \neq n$,
 for any natural numbers n, m .

Functions which meet this additional condition are called *one-to-one*. Another way, then, of thinking of the effect of (2.15) is that it requires that any contextual assignment be one-to-one.⁴

Exercise 2.2

Show in detail how (2.15) successfully blocks (2.14)!

Exercise 2.3

The sentence *She thinks Liv is a spy* does not have a reading on which *she* and *Liv* corefer. Discuss in detail which principle(s) exclude(s) each of the following representations, all of which express the incriminated reading, and which don’t:

- (2.16) (a) She₃ thinks Liv₃ is a spy. (with $g(3) = \text{Liv}$)
 (b) She₃ thinks Liv₂ is a spy. (with $g(3) = g(2) = \text{Liv}$)
 (c) She₃ thinks that Liv is a spy. (with $g(3) = \text{Liv}$)

⁴ It should be noted that assignment functions that aren’t one-to-one can still be used *in the process* of calculating an interpretation; this will be necessary, for example, to interpret a sentence like *Baszon*₃ *believes that everyone*₉ *should get a raise*, which entails that Baszon believes that he himself should get a raise, which, as we will see below, requires an assignment under which both $g(3)$ and $g(9)$ are Baszon. Such an assignment, however, could never be used as a *contextual* assignment by PACO.

Exercise 2.4

It was remarked in section 2.2.2 that indexing first and second person pronouns is necessary to exclude certain BT violations. Provide an example and discuss it!

Exercise 2.5

Is (2.17) a well-formed representation? Should it be? Argue!

(2.17) Latifa₆'s fans admire her₁₄. (with $g(6) = g(14) = \text{Latifa}$)

Exercise 2.6

Above it was said that relative to a context/assignment a sentence containing a pronoun expresses exactly one proposition. But it often happens that you don't quite know which proposition a sentence expresses, precisely because you're not sure who the pronoun refers to. Was the earlier statement wrong, or just sloppy? Clarify!

2.3 Compositional interpretation

In this section, we will provide the formal basis for compositionally interpreting syntactic trees. While all relevant concepts are formally introduced, this section cannot replace a complete introduction to compositional interpretation, for which the interested reader is referred to semantics textbooks (e.g. Heim and Kratzer [1998]:chs. 1–2).

2.3.1 Basics

Types and functions

Above we said that names and NPs in general denote individuals and that sentences denote truth values (both relative to a given assignment). We call those the *denotation domains* of NPs and Ss. Just as expressions belong to syntactic categories, they belong to *semantic types*, according to what kind of denotation domain they denote in. Thus the denotation domain of an NP is the set of individuals (in the technical sense alluded to in section 2.1, including all sorts of objects); we call this set E and we will say that an NP's semantic type is $\langle e \rangle$ (reminiscent of 'entity'). The denotation domain of sentences is the set of truth values, $\{0, 1\}$, and its semantic type shall be $\langle t \rangle$ ('truth value'). Other semantic types and their respective domains can be defined recursively:

- (2.18) Semantic types:
- (a) $\langle e \rangle$ and $\langle t \rangle$ are types
 - (b) if τ_1 and τ_2 are types, then $\langle \tau_1, \tau_2 \rangle$ is a type

(2.19) Denotation domains:

- (a) E (the set of individuals) is the interpretation domain of type $\langle e \rangle$, D_e for short
- (b) $\{0,1\}$ is the interpretation domain of type $\langle t \rangle$, D_t for short
- (c) for any complex type $\langle \tau_1, \tau_2 \rangle$, its interpretation domain D_{τ_1, τ_2} is $D_{\tau_2}^{D_{\tau_1}}$, the set of all functions from D_{τ_1} to D_{τ_2}

Clause (2.19c) introduces the notion of a *function*. What is a function? Take an example: an intransitive verb is of type $\langle e, t \rangle$, i.e. it is a function from individuals to truth values. It ‘takes’ an individual as its *argument*, and ‘gives’ 0 or 1 as its *value*, or, as is usually said, it *maps* an individual onto a truth value. For concreteness take (2.20) (where $f : D_1 \rightarrow D_2$ stands for ‘ f is a function from D_1 to D_2 ’):

(2.20) $\llbracket \text{smiled} \rrbracket^g$ = that function $f : D_e \rightarrow D_t$ such that for all x in D_e , $f(x) = 1$ iff x smiled

According to this, $f(\llbracket \text{Caroline} \rrbracket^g)$ equals 1 if Caroline smiled, 0 otherwise (since there are only two truth values, we can abbreviate this as ‘... 1 iff Caroline smiled’, where ‘iff’ stands for ‘if and only if’); and so on for every other individual. We can now introduce our most important semantic composition rule for non-terminal nodes, *function application* (V^n stands for any verb or projection thereof):

(2.21) Function Application (FA):
 $\llbracket V^n \text{ NP} \rrbracket^g = \llbracket \text{NP } V^n \rrbracket^g = \llbracket V^n \rrbracket^g(\llbracket \text{NP} \rrbracket^g)$

According to this rule, whenever we encounter an NP and a verb or verbal projection, which denotes a function, we *apply* that function to the denotation of the NP. Using lexical meanings and function application, we thus get:

(2.22) (a) $\llbracket \text{Caroline} \rrbracket^g = \text{Caroline}$ (in flesh and blood)
 (b) $\llbracket \text{smiled} \rrbracket^g$ = the function f as defined in (2.20) above
 (c) $\llbracket \text{Caroline smiled} \rrbracket^g = \llbracket \text{smiled} \rrbracket^g(\llbracket \text{Caroline} \rrbracket^g)$ (FA)
 $= f(\text{Caroline}) = 1$ iff Caroline smiled (0 otherwise)

Functions, as in (2.20), are rather cumbersome to write, so it will be useful to introduce some abbreviatory conventions. To name a function like f in (2.20), we will alternatively write (2.23a), or (2.23b):

(2.23) (a) $\lambda x \in D_e. x$ smiled
 (b) $\lambda x_e. x$ smiled

(2.23a) and (2.23b) use the λ (*lambda*)-*prefix*, borrowed again from formal logic. We employ the following convention (cf. Heim and Kratzer [1998]:37):

(2.24) λ -Convention: Read $\lambda\alpha. [\phi \dots \alpha \dots (\alpha) \dots]$ as
 (a) ‘that function which maps every a to 1 iff $[\phi \dots a \dots (a) \dots]$ ’, if ϕ is a sentence,
 (b) ‘that function which maps every a to $[\phi \dots a \dots (a) \dots]$ ’ otherwise

We will return to clause (2.24b) below. Since ‘ x smiled’ in (2.23) is a sentence, clause (2.24a) applies: $[\lambda x_e.x \text{ smiled}]$ is that function which maps every x in D_e to 1 iff x smiled (brackets added for perspicuity). Note that the type of a λ -expression can be read off it straightforwardly: (2.23a)/(2.23b) for example has a λ binding a variable of type $\langle e \rangle$, followed by a sentence, ‘ x smiled’. English sentences represent meanings in D_t , which means that the whole expression is of type $\langle e, t \rangle$.⁵

Transitive verbs

A typical transitive verb denotation is given in (2.25):

$$(2.25) \quad \llbracket \text{likes} \rrbracket^g = \lambda x \in D_e. \lambda y \in D_e. y \text{ likes } x$$

Here the sentence $y \text{ likes } x$ has two λ s in front of it, each binding an individual variable. Its semantic type is thus $\langle e, \langle e, t \rangle \rangle$, or for short: $\langle e, et \rangle$. To disentangle this, we first apply (2.24b) with $\alpha = x$ and $\phi = [\lambda y \in D_e. y \text{ likes } x]$, whereby (2.25) is a function which maps every individual x onto $[\lambda y \in D_e. y \text{ likes } x]$. The latter in turn, according to (2.24a), is a function that maps an individual y to 1 iff y likes x . (2.25) as a whole, then, is a function from individuals to functions from individuals to truth values. Such a function subsequently takes two individuals and maps them onto a truth value:

$$(2.26) \quad \begin{aligned} (a) & \quad \llbracket \text{Chris likes Caroline} \rrbracket^g = \\ (b) & \quad \llbracket \text{likes Caroline} \rrbracket^g (\llbracket \text{Chris} \rrbracket^g) = \\ (c) & \quad \llbracket \text{likes Caroline} \rrbracket^g (\text{Chris}) = \\ (d) & \quad \llbracket \text{likes} \rrbracket^g (\llbracket \text{Caroline} \rrbracket^g) (\text{Chris}) = \\ (e) & \quad \llbracket \text{likes} \rrbracket^g (\text{Caroline}) (\text{Chris}) = \\ (f) & \quad [\lambda x \in D_e. \lambda y \in D_e. y \text{ likes } x] (\text{Caroline}) (\text{Chris}) = 1 \text{ iff Chris likes Caroline} \end{aligned}$$

Functions and the sets they characterize

At the beginning of this section we said that an intransitive verb denotes a function from individuals to truth values (type $\langle et \rangle$). In section 2.1, however, it was said that an intransitive verb denotes a set of individuals. While only the former is literally true, the latter almost is, because any function of type $\langle \tau, t \rangle$ uniquely characterizes a set (namely a subset of D_τ), and vice versa. The function $[\lambda x_e.x \text{ smiled}]$ – call it f_{smiled} – for example, characterizes the set of all individuals x which that function maps to 1, formally: $\{x \in D_e \mid f_{\text{smiled}}(x) = 1\}$; informally: the set of all those who smiled.

In the remainder of this book, I will often make use of this equivalence, which is rendered again in (2.27):

⁵ Note that individuals and functions *are* the meanings of object language expressions. We use names, English sentences, and λ -expressions to *name* those denotations in our meta-language; accordingly, no italics are used here. I will, however, sometimes include expressions used to name a denotation in quotation marks to facilitate reading.

- (2.27) Set/function equivalence and convention:
 ‘ $[\lambda x \in D_\tau \dots x \dots](y)$ ’ is equivalent to ‘ y is in $\{x \in D_\tau \mid \dots x \dots\}$ ’; we will often write the latter instead of the former

For example, ‘ $[\lambda x \in D_e . x \text{ smiled}](\text{Caroline}) = 1$ ’ is equivalent to ‘Caroline is in $\{x \in D_e \mid x \text{ smiled}\}$ ’. The latter form is sometimes more convenient and more transparent, which is why I will use it. Also, I find it more intuitive to think of an intransitive verb denotation as a set of individuals, and of a sentence containing it as true if the denotation of the subject is in that set, than to think of an intransitive verb denotation as a function from individuals to truth values, and of a sentence containing it as true if that function applied to the subject denotation is 1 – though, as stated, both are equivalent.

So, to reiterate, I will regularly describe a denotation by saying ‘so-and-so is in set- x ’ where our official semantics assign it the denotation ‘function- x applied to so-and-so = 1’.

The reason why we still officially use ‘function talk’ rather than ‘set talk’ is that the former is more general. Recall that, for example, a sentence with a transitive verb can be interpreted by repeated function application, as in (2.26), because a transitive verb denotes a function whose values are again functions. While, alternatively, we could think of a transitive verb as denoting pairs of individuals – those pairs $\langle x, y \rangle$ in which x loves, hates, knows, calls, etc. y – such a meaning would require an extra rule for combining transitive verb meanings with object NP meanings.

Compositionality

A theory of interpretation should be *compositional*, meaning: denotations of complex expressions should be composed of the denotations of the simpler expressions that make them up. For concreteness, let us adopt the following two conditions on interpretations:

- (2.28) (a) Every syntactic constituent C has an interpretation.
 (b) In interpreting a constituent C , no other information can be used for computing $\llbracket C \rrbracket^g$ than that associated with C ’s daughters.

The Function Application rule used so far is strictly compositional in this sense. Consider, for illustration, a different rule:

- (2.29) Non-Compositional S-rule:

$$\left[\begin{array}{c} \text{S} \\ \swarrow \quad \searrow \\ \text{NP}_S \quad \text{VP} \\ \quad \swarrow \searrow \\ \quad \text{V} \quad \text{NP}_O \end{array} \right]^g = [\text{V}]^g([\text{NP}_O]^g)([\text{NP}_S]^g)$$

This rule will assign the same denotations to a transitive sentence as the repeated application of FA. It is, however, non-compositional. For one thing, VP doesn’t

Table 2.1 *Types and their names*

Type	Name	Variable	Categories
e	individual	x, y, z	names, pronouns
t	truth value	p	sentences
et	properties	P	VPs, Ns, APs
e,et	relations	R	transitive Vs and Ns
et,t	generalized quantifier	q	quantified NPs

receive an interpretation, *contra* (2.28a). And, for another, the meaning rule for S directly uses the meanings of V and NP_O , which are not daughters of S, in violation of (2.28b). We will continuously check that all our interpretation rules to follow are compositional in nature.

A first inventory of categories and types

Different semantic types have common names in the semantic literature, which are summarized in table 2.1 (not all of them will look familiar to you yet).

In table 2.1 I also indicate the typical variables I use for the different semantic types. This is, of course, just a notational convention, but it will allow us to abbreviate the function notation further by leaving out the explicit domain specification. Instead of (2.25) we can now write (2.30), because it is understood that x and y are variables over D_e (upon demand, variables will be numbered, e.g. $p_1, p_2, \dots$):

$$(2.30) \quad \llbracket \text{likes} \rrbracket^g = \lambda x \lambda y. y \text{ likes } x$$

Exercise 2.7

Why must the denotation for, say, *likes* be as in (2.25), rather than $[\lambda y \in D_e. \lambda x \in D_e. y \text{ likes } x]$? What is the difference? Argue!

Exercise 2.8

Give the denotation for the verb *danced* both in set notation and λ -notation.

Exercise 2.9

Give the denotation for the verb *introduced*.

Exercise 2.10

Give denotations for all lexical elements in the following sentences, and provide derivations of their truth conditions à la (2.26); add indices where necessary:

- (2.31) (a) Sam danced.
 (b) Kim likes Jo.
 (c) She slept.
 (d) He admires him.
 (e) He admires himself.

2.3.2 Truth, truth conditions, and meanings

The denotation of a sentence is a truth value, 1 or 0. What is closer to the intuitive meaning of a sentence, however, is not its truth value, but the *truth conditions* the interpretation procedure assigns to it, for example a statement of the form ‘ $\llbracket S \rrbracket = 1$ iff ...’. The goal of semantic theory, after all, cannot be to determine if a sentence is true or false (unless it happens to be a tautologous or contradictory sentence), but to assign systematically to every sentence a description (in English or something resembling English) of the kind of situation in which it would be true.

A slightly different way to think of the general meaning of a sentence involves the situations a sentence correctly describes. Generally, the following holds:

- (2.32) Given truth conditions of the form ‘ $\llbracket S \rrbracket = 1$ iff ϕ ,’ we can say that S is true in a situation s iff ϕ holds in s .

Thus from the truth conditions ‘ $\llbracket \text{Caroline smiled} \rrbracket^g = 1$ iff Caroline smiled’ we can go to ‘*Caroline smiled* is true in a situation s if Caroline smiled in s .’ The set of situations in which a sentence S is true is often called *the proposition expressed by S* . Instead of saying that a sentence denotes 1 iff ϕ , we will sometimes just say that it denotes the proposition that ϕ ; thus *Caroline smiled* denotes the proposition that Caroline smiled.

But sentences contain NPs, and NPs depend on the assignment function, so we have to refine this. Recall that the denotation of an NP_n is defined relative to an assignment g , if $g(n)$ meets the presuppositions of the denotation of NP (e.g. that it be a female individual, or that it be called John, etc.); otherwise $\llbracket \text{NP}_n \rrbracket^g$ is *undefined*. We can thus say the following:

- (2.33) Given truth conditions of the form ‘ $\llbracket S \rrbracket^g = 1$ iff ϕ ,’ we can say that S is true in a situation s relative to assignment g if $\llbracket S \rrbracket^g$ is defined and ϕ holds in s (it is false if $\llbracket S \rrbracket^g$ is defined and ϕ doesn’t hold in s).

Whether or not $\llbracket S \rrbracket^g$ is defined for a given S thus depends on what assignment g we choose. The *proposition expressed by S relative to assignment g* is the set of situations in which S is true relative to g . A given sentence can thus express different propositions relative to different assignments.

This is plausible if, again, we think of an assignment as (part of) a context. Take as a concrete example the sentence *She₆ smiled*. We derive the truth conditions $\llbracket \text{she}_6 \text{ smiled} \rrbracket^g = 1$ iff $(g(6)$ is a female individual and) $g(6)$ smiled. By (2.33), it then holds that: for any assignment g , $\llbracket \text{she}_6 \text{ smiled} \rrbracket^g$ is defined iff $g(6)$ is a single female individual; if defined, it denotes the proposition that

$g(6)$ smiled. Take a g such that $g(6) = \text{Caroline}$; then that means that ‘Caroline smiled’ is a possible meaning of she_6 smiled. But given that there are other assignments g' such that $g'(6) = \text{Susan}$, ‘Susan smiled’ is also a possible meaning of she_6 smiled, and so on and so forth, for all and only those assignments according to which $g(6)$ is single and female.

Obviously, we haven’t said anything about how g relates to the actual linguistic context, in other words: about the question which individual a given pronoun will, or is likely to, denote in a given linguistic context (see subsection 2.4.1 for a few remarks). Yet, by virtue of universally quantifying over all technically possible g , we can define the context-independent notion of a possible meaning:

- (2.34) ϕ is a *possible meaning* of a sentence S iff there is an assignment function g such that $\llbracket S \rrbracket^g = 1$ iff ϕ

For example, ‘Caroline smiled’ is a possible meaning of She_6 smiled, as is ‘Susan smiled’. ‘Bill smiled’ or ‘Lake Balaton smiled’ are not, assuming that neither Bill nor the popular Hungarian lake are female individuals. More interestingly, given PACO in (2.15), we derive for example that ‘Caroline likes Susan’ is a possible meaning of $Caroline_2$ likes her_3 , but ‘Caroline likes Caroline’ is not.

Note that the possible meanings for a sentence are independent of the actual choice of indices. Two sentences S_1 and S_2 that only differ in indexing have the same set of possible meanings, provided all and only those NPs in S_2 are coindexed that are coindexed in S_1 . For example, the set of possible meanings of She_n smiled is the same for any choice of index n .⁶

As a consequence, we can go one step further and define the notion *possible meaning of an expression* E , where an expression is a sentence (or rather: tree) without indices: if ϕ is a possible meaning of E , that means that there is some indexing E^I of E which is grammatical (i.e. obeys the Binding Conditions A–C and whatever other syntactic conditions we’ll impose on indexing), and ϕ is a possible meaning of that indexed version E^I of E .

For example, let E be the expression *She likes her*, which has an infinite number of possible indexings E^I . By Binding Condition B, all of these indexings must assign different integers to *she* and *her*. By the PACO, such an indexing prohibits an interpretation by which *she* and *her* wind up coreferential. Thus the set of possible meanings for *She saw her* (without indices) is the set of all propositions ‘ x saw y ’ where x and y are two different female individuals.

The expression *She says Kim likes her*, on the other hand, has infinitely many indexings on which *she* and *her* are coindexed, and infinitely many on which they are not (and all of those have *Kim* counter-indexed with both pronouns). The former all get interpretations of the form x says *Kim likes* x , the latter of the

⁶ This is not to say that there couldn’t be *syntactic* rules or constraints that distinguish S_1 and S_2 , as argued recently in e.g. Heim (1997), or Sauerland (2000); we will discuss some relevant cases in sections 5.5.2 of chapter 5, and 6.5.1 of chapter 6.

Table 2.2 *Semantic objects related to sentences*

truth value	0 or 1
truth conditions	$\llbracket S \rrbracket^g = 1$ iff ϕ
proposition expressed by S (relative to g)	ϕ / the set of situations s in which ϕ holds (assuming that $\llbracket S \rrbracket^g = 1$ iff ϕ)
possible meanings of indexed S	the set of propositions S can express relative to any g (respecting PACO)
possible meaning of unindexed S	the set of propositions any grammatical indexing of S can express relative to any g (respecting PACO)

form x says *Kim likes y*, with $x \neq y \neq Kim$. Hence all instances of these are possible meanings for the expression *She says Kim likes her*.

In conclusion, the various semantic objects corresponding roughly to whole sentences introduced in this subsection are summarized in table 2.2. Since natural language doesn't come with indices, possible meanings of expressions/unindexed sentences are what our semantic interpretation procedure, in tandem with Binding Theory, should ultimately predict.

Exercise 2.11

Give truth conditions, the proposition expressed, and, where you can, the possible meanings (regardless of indexing) for the sentences in (2.31) (again, add indices where necessary).

Appendix: Meaning relations

Given the above, we define, for the sake of completeness, some essential semantic notions for indexed sentences:

- (2.35)
- (a) $S1$ entails $S2$ iff for any assignment g , every situation in which $S1$ is true relative to g is one in which $S2$ is true relative to g .
 - (b) S is a *tautology* iff it is true in any situation relative to any assignment (for which it is defined).
 - (c) $S1$ and $S2$ are *incompatible* if for any assignment g , there is no situation in which $S1$ and $S2$ are both true relative to g .
 - (d) $S1$ and $S2$ are *synonymous* if each entails the other.

2.3.3 More composition rules [Ⓔ]

Type-driven interpretation

In (2.26) above we used Function Application twice, once to combine the object meaning with the transitive-verb meaning, and then again to combine that VP meaning with the subject meaning. Note that in each case, FA is well defined, since the type of the argument, $\langle e \rangle$, 'matches' the type of the function,

$\langle e, \dots \rangle$. This is so because any element of type $\langle e, \dots \rangle$ denotes a function from individuals, i.e. elements of D_e , to something else (depending on what $\dots$ is). In a case like that we say that the argument, here an individual in D_e , is in the *domain* of the function.⁷

Among two given meanings M_1 and M_2 , exactly one of three possible situations will obtain:

- M_1 is in the domain of M_2 , i.e. $M_2(M_1)$ is defined
(e.g. M_1 is in D_e and M_2 is in $D_{e,t}$ or $D_{e,et}$)
- M_2 is in the domain of M_1 , i.e. $M_1(M_2)$ is defined
(e.g. M_2 is in D_e and M_1 is in $D_{e,t}$ or $D_{e,et}$)
- neither is in the domain of the other, i.e. neither $M_1(M_2)$ nor $M_2(M_1)$ are defined
(e.g. both M_1 and M_2 are in D_{et})

Given this, we can generalize our interpretation rule as follows:

- (2.36) Function Application, revised and final:
 $\llbracket A B \rrbracket^g = \llbracket A \rrbracket^g(\llbracket B \rrbracket^g)$ or $\llbracket B \rrbracket^g(\llbracket A \rrbracket^g)$, whichever is defined

The reader may verify that this rule subsumes our earlier (2.21). An interpretation procedure along these lines, where the semantic type of the sister constituents (rather than their syntactic category and/or linear order) determines the mode of semantic composition, is called *type-driven interpretation* or, less respectfully, ‘shake-and-bake semantics’; it was introduced into linguistic semantics in Klein and Sag (1985).

We will add a few other composition rules that take care of situations in which neither $M_1(M_2)$ nor $M_2(M_1)$ in the sense above are defined later on, but FA alone will bring us a long way. For concreteness, we will assume that a constituent to which no semantic composition rule applies is uninterpretable and hence ungrammatical.

Interpreting flat structures

In chapter 1 we encountered examples which appear to involve flat structures, i.e. nodes with more than two daughters, e.g. double object VPs, NPs with two post-nominal PPs, and generally clausal structures in non-configurational languages:

- (2.37) (a) I introduced the students to each other.
 (b) (Ja čital) stat’ju Tolstoja o sebe. (Russian)
I read article-ACC T.-GEN about himself-LOC
 ‘(I read) an article of Tolstoj about himself.’

⁷ Confusion is lurking here: note that the *denotation domain* of an expression E is not identical to the *domain* of the *denotation* of E , even if E denotes a function. For example, the denotation domain of an intransitive verb is D_{et} , the set of functions from individuals to truth values, but the domain of the denotation of an intransitive verb is D_e , the set of individuals.

We suggested there that binding asymmetries among the daughters in such a multiply branching structure are encoded in a non-phrase structural aspect of syntactic representation, say the a(rgument)-list on which a command relation such as O(bliqueness)-Command can be defined. But, independent of the proper formulation of the Binding Conditions, we also need to say something about the interpretation of these structures, as our Function Application rule (2.36) (or its earlier incarnation [2.21]) obviously doesn't apply to such a structure. This can be done by adding a rule like the following:

(2.38) *N-ary Function Application:*

$$\left[\left[\begin{array}{c} X^n \\ \swarrow \quad \downarrow \quad \searrow \\ X^0 \quad A \quad A' \quad A'' \dots \end{array} \right] \right]^g = \llbracket X^0 \rrbracket^g (\llbracket A_n \rrbracket^g) (\llbracket A_{n-1} \rrbracket^g) \dots (\llbracket A_1 \rrbracket^g)$$

where $A_1, A_2, \dots, A_{n-1}, A_n$ is the order of $A, A', A'' \dots$ on X^0 's argument-list

For example, *introduce Helena to Gwynn* will be interpreted as $\llbracket \text{introduce} \rrbracket^g (\llbracket (\text{to}) \text{ Gwynn} \rrbracket^g) (\llbracket \text{Helena} \rrbracket^g)$, provided that the a-list of *introduce* looks like $\langle \text{NP}, \text{NP}_{\text{acc}}, \text{PP} \rangle$.⁸ Note that this rule is strictly compositional, since it assigns a meaning to every node, based on the meanings of its daughters.

2.4 Extensions and alternatives^⑤

2.4.1 A note on non-C-command anaphora

Above, we somewhat boldly called g (part of the representation of) the 'context'. However, nothing in our theory tells us how g relates to an actual context. Consider, for example, the following:

- (2.39) (a) A dog is sitting at the bar. (b) Suddenly, it sees a man come in. (c) The man has a dog too. (d) It is wearing a tie.

Here's a selection of what our treatment so far tells us: (i) *the bar* cannot be bound by *a dog* in (a); (ii) *it* in (b) can refer to the dog from (a), or some other neuter individual; (iii) *the man* in (c) refers to the unique man, which can be the man from (b); etc. etc. Here's what our treatment doesn't tell us, though native speakers could:

⁸ Notice that the FA notation is left-associative, i.e. $X(A)(A')(A'')$ is equivalent to $((X(A))(A'))(A'')$. In other words, semantic composition still proceeds 'binary,' combining with one argument 'at a time.' We will justify this approach (rather than, say, one which combines the verb meaning with a tuple consisting of all its argument meanings at once) in chapter 4 below.

Note, too, that (2.38) is built on the assumption that all but the least oblique argument A_0 on the a-list are sisters to X^0 , though this could easily be refined.

- (2.40)
- (a) *it* in (b) is very likely to be taken to refer to the dog mentioned in (a), rather than to any other individual
 - (b) *it* in (b) cannot possibly refer to the other dog (the one with the tie)
 - (c) *the man* in (c) is very likely to be interpreted as the man mentioned in (b)
 - (d) *a dog* in (c) cannot be the dog mentioned in (a) and (b)
 - (e) *it* in (d) is very likely to be taken to refer to the second dog

In short, what is missing is a theory about how a given sentence *changes* the context, including the referential options for sentences to follow. How do referents become available? How do speakers choose among different grammatically possible antecedents for a given pronoun?

Informally, such a theory would start from the assumption that at the beginning of a discourse there are no anaphoric possibilities; technically, a discourse starts with an assignment function with an empty domain. NP occurrences can then be classified into those that *introduce discourse referents*, hence add their index to the domain of the assignment function, and those that don't, hence rely on their index being in the domain of the assignment function already; the latter are the anaphoric NPs.

Whether or not an NP belongs to the introducing or the anaphoric type is not merely a matter of its morphological shape. Proper names, for example, can be used anaphorically, or to introduce a new discourse referent; so can definite NPs (*the moon* versus *the guy*), demonstratives, and first and second person pronouns (though the latter could be argued to be 'anaphoric' by definition). Even third person pronouns, though most commonly used anaphorically, can be used without an antecedent, for example when using, say, *she* to refer to a person who just walked in, or whom I point to, or whom we both watch on a TV monitor (though one could argue that their very appearance, or my pointing at them, actually introduces the discourse referent, which is then anaphorically picked up by the use of the pronoun). Indefinite NPs, including cardinals and certain quantifiers such as *few*, *most*, etc. are perhaps the only morphological class that is consistent, in that they always introduce new referents, and cannot, it seems, be used anaphorically.

Discourse Representation Theory provides a framework in which notions such as 'having a discourse antecedent,' 'introducing a discourse referent,' or 'being incapable of having a discourse antecedent' can be formalized; representative examples are Heim's (1982, 1983) *File Change Semantics*; Kamp's (1981) *Discourse Representation Theory*; and Groenendijk and Stokhof's (1991) *Dynamic Predicate Logic*. Useful introductory texts include Kamp and Reyle (1993) and Chierchia (1995).

In addition, a theory of discourse anaphora should address the question which type of NP will most likely be used anaphorically to pick up a particular referent (it seems that the choice between, say, a pronoun and a definite NP depends on roughly how long ago the pertinent discourse referent was introduced), and conversely, which one of a number of already introduced referents a given anaphoric NP is likely to pick out. Theories that address these questions include *Centering*

Theory (Grosz *et al.*, 1995; Walker *et al.*, 1998), *Accessibility Theory* (Ariel, 1999, 2001), and *Relevance Theory* (Sperber and Wilson, 1995) among many others; see also Kehler (2002) for a hybrid proposal.

I will not introduce a theory of context change, centering, or relevance in this book, which focuses mainly on binding within a sentence; that is, as far as our official theory goes, (b) in (2.39) for example, has as its possible meaning ‘Fido sees a man come in,’ where Fido is the dog mentioned in (a), but also ‘Hasso sees a man come in,’ where Hasso is the other dog mentioned in (c), and that it is up to a discourse theory to filter further among these.

It is worth noting that the question when an NP can pick up the referent introduced by another NP’ can sometimes influence our acceptability judgments on a simple sentence as well. For example, most speakers will find it very hard to interpret *his* and *Edson* as coreferential in (2.41a), while that interpretation is easy to get in (2.41b):

- (2.41) (a) How did other mothers cope with this? For example, what did *his* mother think, when she learned *Edson* got arrested?
 (b) How did other mothers cope with this? For example, what did *Edson’s* mother think, when she learned *he* got arrested?

The same sentence as in (2.41a), however, improves considerably on a coreferential reading if the preceding context is changed:

- (2.42) What did his family think of *Edson*? For example, what did *his* mother think, when she learned *Edson* got arrested?

This pattern suggests that the problem with (2.41a) is that *his* cannot pick up the referent introduced by *Edson*, presumably because it precedes it (whence the improvement if the NPs are swapped as in [2.41b]). If the previous sentence already talks about *Edson* as in (2.42), no such ‘backwards link’ is necessary, and *his* is naturally understood to refer to *Edson*. It seems appropriate to say that while *his* and *Edson* corefer in the second clause in (2.42), *his* is actually anaphoric to the occurrence of *Edson* in the first sentence in (2.42) (as is the second occurrence of *Edson*, presumably).

To be sure, these remarks are not meant to provide a serious account of so-called *backwards anaphora*; it is quite possible that in some configurations, pronouns can be genuinely anaphoric to a full NP that *follows* them. If our remarks about the above examples are on the right track though, they do suggest that we were justified in separating out the purely structural Binding Conditions A–C, which have nothing to say about any of these cases, from a more general theory of introducing and anaphorically picking up discourse referents, which applies to intra- and inter-sentential anaphoric relations alike.⁹

⁹ This doesn’t mean that there can’t be controversy about which kind of effect belongs to which theory. Evans (1980) claims that Binding Condition C effects are not structural but result from

2.4.2 Reflexives as reflexivizers

Above we assumed that reflexive and non-reflexive pronouns are semantically identical; they are interpreted like individual variables. Whatever differences we find between them are accounted for in the syntax, i.e. by indexing. An alternative approach is to view reflexives as *semantically* distinct from non-reflexive pronouns. As a starting-point, assume that the reflexive in a sentence like *Natasha painted herself* serves to map the relation denoted by *painted* onto the property of self-painting. Formally, this means that the meaning of *herself*, represented as *SELF*, maps a relation *R* onto the property of being an individual that stands in the *R* relation to itself, $\lambda x.R(x)(x)$:

- (2.43) (a) *SELF* is that function from relations to properties such that $SELF(R) = \lambda x.R(x)(x)$ for all $R \in D_{e,et}$
 (b) $\llbracket \text{himself/herself/itself} \rrbracket^g = \lambda R \lambda x.(x \text{ is a male/female/neuter individual and}) SELF(R)(x)$

Interestingly, a reflexive on this view does not need to be indexed, since it is not a referring expression. The fact that it is ‘locally bound’ simply follows from its lexical meaning: it maps the predicate it combines with onto what we might call a ‘reflexive property.’ Thus something close to Binding Condition A actually follows from this analysis of reflexives.

Obviously, nothing about non-reflexives follows. One could formulate principles to re-establish complementary distribution again, however, e.g. (2.44):

- (2.44) (a) No pronominal can be used where a reflexive would yield the same meaning.
 (b) Unless a transitive verb *V* has a reflexive pronoun as its argument, interpret it as $\lambda x.\lambda y.\llbracket V \rrbracket^g(x)(y) \ \& \ x \neq y$.

As just discussed, this approach is attractive in that it builds Binding Condition A into the lexical meaning of the reflexive: the reflexive will automatically be ‘bound,’ and it will be bound by a higher coargument. It is, however, well known that reflexives do not always combine with a relation denoting expression, cf. (2.45a) and (2.45b), and that in such cases the binder of a reflexive is not always the next higher coargument (in [2.45a] the ‘highest’ argument, the subject, binds the ‘lowest,’ the direct object, skipping, as it were, the indirect

the fact that a pronoun is trying to pick up its referent from a full NP it c-commands (he claims, in line with the other authors quoted at the end of section 1.2.3 in chapter 1, that a full NP c-commanding a coreferent full NP isn’t ungrammatical at all). He adduces the acceptability of (i) below (his [59]) as evidence for that position. I believe though that (i) instantiates a special case of what Heim (1993) calls ‘when structured meanings matter’, which we will discuss briefly at the end of section 6.3 in chapter 6. Generally, a pronoun c-commanding a coreferent NP, even in the presence of an independent antecedent as in (ii), appears to be rather bad (compare [ii] to [2.42]), and should be ruled out on structural grounds:

- (i) What do you mean *John* loves no one? *He* loves *John*.
 (ii) *I wonder how *Edson* felt. What did *he* think when *Edson* got arrested?

object in the middle); indeed, the antecedent needn't be a coargument at all, cf. (2.45c) and (2.45d):

- (2.45) (a) Gabi introduced herself to John.
 (b) Gilbert₁ told Spencer₂ about himself_{1/2}.
 (c) The president saw [himself give a speech].
 (d) The CEO despised [those recent articles about herself].

In none of these cases can the reflexive be interpreted as in (2.43b); furthermore, to capture the two coreference options in (2.45b), we would have to assume two different meanings for the reflexive (or two different structures for the sentence). While appropriate meanings can be devised, the challenge is to show that the resulting class of meanings is constrained in some natural way. Otherwise, the distribution of reflexives reduces to an idiosyncratic case of lexical ambiguities, hardly a satisfactory state of affairs.¹⁰

On the other hand, even if this treatment turns out to be inadequate for English *himself*, there might be other elements in other languages that are amenable to it. In particular, many languages have a verbal reflexive marker which essentially de-transitivizes a transitive verb to give you an intransitive verb. This marker might aptly be characterized as denoting SELF. Consider the following examples from Finnish and Chicheŵa (from Sells *et al.* [1987]:177,187):

- (2.46) (a) Jussi pese -yty -i. (Finnish)
J. washed self PAST
 'Jussi washed himself/self-washed.'
 (b) Jussi puolusta -utu -i.
J. defended self PAST
 'Jussi defended himself/self-defended.'
- (2.47) Alenje a- na- dzi- lum -a. (Chicheŵa)
hunters SA PAST self bite INDICATIVE
 'The hunters bit themselves/self-bit.'

The affixes *-yty/-utu* and *dzi-* are not syntactic arguments or clitics (they cannot, for example, be separated from the verb or be coordinated with other NPs; see Mcahombo [1993a] for more evidence), and the derived form is syntactically intransitive (e.g. the *-yty/-utu* forms form causatives following the intransitive instead of the transitive pattern). It seems plausible that these affixes semantically function like SELF described above. Likewise, certain other reflexives with a more limited distribution than English *himself* are perhaps best treated this way (cf. Sells *et al.* [1987]).

Exercise 2.12

The relation that is reflexivized in (2.45a) can be written as $\lambda x.\lambda y.Gabi\ introduced\ y\ to\ x$. (2.48) gives the appropriate lexical entry for

¹⁰ Another problem with this approach is what happens if the reflexive is coordinated, as in *Polsen painted himself and Mary*, especially on a *non-distributive* reading of the coordination, a matter we cannot go into here.

herself to derive this meaning (where R^3 is a variable over three-place relations, i.e. a variable in $D_{e,\langle e,et \rangle}$):

$$(2.48) \quad \llbracket \text{herself}_{dt1} \rrbracket^g = \lambda R^3. \lambda x_1. \lambda x_2. R^3(x_1)(x_1)(x_2) \\ (= \lambda R^3. \lambda x_1. \lambda x_2. \text{SELF}(\lambda x_3. \lambda x_3. R^3(x_3)(x_4)(x_2))(x_1))$$

Note that this rule assumes that the order of Function Application for $V NP_1$ [to NP_2] is actually $\llbracket V \rrbracket^g (\llbracket NP_2 \rrbracket^g) (\llbracket NP_1 \rrbracket^g)$.

Characterize the relation that is reflexivized in (2.45b) in the same way. Then give a lexical meaning for *herself* that derives the correct meaning.

Exercise 2.13

Consider the following Kannada data (from Lidz [1995]):

- (2.49) (a) shyaamu₁ raamu₂ tann-annu_{1/*2} hoDe -d -a
 S. R. PRON-ACC hit -PAST-3SM
 anta heeL -id -a
 that say PAST 3SM
 ‘Shyamu said that Raamu hit him.’
- (b) shyaamu₁ raamu₂ tann-annu_{2/*1} hoDe -du -koND
 S. R. PRON-ACC hit -PAST REFL.
 -a anta heeL -id -a
 -3SM *that say PAST 3SM*
 ‘Shyamu said that Raamu hit himself.’

What unexpected property do these sentences show? Discuss! Does this suggest alternative treatments for the Finnish and Chicheŵa cases above?